AI-Powered Kitchen Ecosystem for Food Waste Reduction and Nutrition Optimization

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***Abstract*— Household food management remains a persistent global challenge, contributing significantly to food waste and nutritional imbalance. Existing digital solutions often operate in isolation and lack cultural adaptability, multilingual accessibility, and integrated decision support across the full food lifecycle. This study presents an Artificial Intelligence Powered Kitchen Ecosystem, a unified framework that integrates semantic recipe intelligence, predictive nutritional risk classification, safety-constrained food expiry estimation, and hybrid shopping recommendation within a single modular architecture. The cooking assistant achieved 86.84 percent classification accuracy and demonstrated a statistically significant 73.9 percent reduction in weekly household food waste during a three-week user study. The nutritional guidance module achieved 88.12 percent classification accuracy with strong multi-class separability. The personalized expiry prediction model explained 74.43 percent of variance with an average deviation of approximately 2.3 hours. The hybrid recommendation engine achieved high ranking performance across multiple evaluation metrics. The proposed ecosystem demonstrates the effectiveness of integrated, safety-aware, and culturally adaptive artificial intelligence systems in improving sustainable household food management.**

Keywords— Food Waste Reduction, Artificial Intelligence, Machine Learning, Nutritional Risk Prediction, Shelf-Life Prediction, Recommender Systems, Multilingual Natural Language Processing

I. Introduction

The modern household kitchen is a nexus of intent and reality, where the goals of healthy eating, economic prudence, and environmental responsibility often collide with the pressures of daily life. This collision manifests as a persistent global challenge: inefficient food management. Annually, approximately one-third of all food produced for human consumption is lost or wasted, placing an immense strain on both household economies and the environment [1]. Concurrently, nutritional deficiencies remain a significant public health concern, often stemming from a lack of personalized dietary awareness and planning [2]. While these issues are interconnected, the digital tools designed to address them typically exist in isolation, creating a fragmented experience for the user.

Current technological solutions, such as recipe applications, nutrition trackers, or smart shopping lists, often operate in silos. A user might consult a recipe app for dinner, a separate app to log into its nutritional content, and a handwritten list for groceries, all while trying to mentally track the expiry dates of items in their pantry. This disjointed workflow lacks the interoperability and contextual awareness needed to drive meaningful behavioral change. Furthermore, most platforms are developed with a Western-centric bias, failing to accommodate the linguistic diversity, unique ingredients, and cultural cooking practices that define cuisines in regions like South Asia [3]. This gap between generic technology and specific cultural contexts limits the adoption and effectiveness of digital food management solutions.

To bridge this critical gap, this paper introduces the design, implementation, and validation of an AI-Powered Kitchen Ecosystem. This is a novel, integrated framework that orchestrates the entire household food lifecycle, from proactive planning and intelligent purchasing to safe consumption and health monitoring. The ecosystem

synergistically combines four core, intelligent modules: a Spontaneous Cooking Assistant that helps users discover recipes based on available ingredients; a Nutritional Guidance system that provides personalized deficiency risk predictions; a Personalized Food Expiry predictor to minimize spoilage; and a Smart Shopping Agent that generates context-aware grocery lists. A foundational principle of our design is deep cultural and linguistic integration, specifically supporting English, Sinhala, and Tamil, to ensure accessibility and relevance for the target population.

The primary contributions of this work are fivefold:

1. The design and implementation of a holistic, multi-modal AI ecosystem that unifies recipe discovery, nutritional analysis, expiry management, and shopping assistance into a single, coherent platform.
2. A novel hybrid recipe retrieval system within the Cooking Assistant that combines contextual semantic embeddings (Sentence-BERT) with culturally aware fuzzy ingredient matching to overcome lexical variation.
3. A safety-constrained expiry prediction model that integrates data-driven forecasts using CatBoost with hard-coded safety bounds, ensuring user safety over predictive accuracy.
4. A personalized shopping recommendation engine that employs a hybrid collaborative filtering and content-based approach, enhanced by trilingual semantic similarity, to suggest relevant and healthy items.
5. Comprehensive empirical validation of the entire system, including a three-week real-world user study (n=15) that demonstrates a statistically significant and practically impactful 73.9% reduction in household food waste.

The remainder of this paper is structured as follows. Section II reviews related work in the domains of recipe recommendation, nutrition tracking, and food waste reduction. Section III presents the overall system architecture and the interaction between its modules. Section IV details the methodology, including dataset construction, feature engineering, and machine learning models for each component. Section V provides a comprehensive evaluation of the system's performance, including the results of the user's study. Finally, Section VI concludes the paper and outlines directions for future research.

II. Literature Review

The challenge of intelligent food management has generated substantial research across food computing, recommender systems, and digital health domains. However, these advancements have largely progressed independently, resulting in specialized yet fragmented solutions. The following review synthesizes key developments across recipe intelligence, nutritional modeling, expiry prediction, and grocery recommendation, highlighting their limitations and motivating the need for integrated ecosystem-level architectures.

A. Recipe Recommendation and Cooking Assistance

Recipe recommendation systems initially relied on keyword matching and TF-IDF vectorization, which captured lexical similarity but lacked contextual understanding. The introduction of distributed embeddings and later transformer-based encoders such as Sentence-BERT significantly improved semantic retrieval performance [12]. However, even advanced embedding models remain sensitive to lexical and cultural variation in ingredient naming, particularly in multilingual contexts. Furthermore, widely used datasets such as Food-101 predominantly represent Western cuisine [14], limiting generalizability to culturally diverse regions. Fuzzy lexical similarity measures such as Levenshtein distance [17] are therefore often required to address naming inconsistencies. These limitations highlight the necessity of hybrid retrieval architectures that integrate contextual semantic embeddings with culturally aware fuzzy matching to achieve robust and culturally adaptive recipe recommendations.

B. Nutritional Guidance and Health Monitoring

Digital nutritional platforms have evolved from basic calorie tracking tools to machine learning–based health monitoring systems. Ensemble methods such as gradient boosting and Random Forest classifiers [7], [8] have demonstrated strong predictive capability in structured health datasets. However, many existing systems rely primarily on raw nutrient intake totals without normalization against standardized dietary reference guidelines [26], [27], limiting interpretability across age groups and physiological conditions. Furthermore, most nutritional monitoring tools function as standalone applications, operating independently of meal planning, inventory awareness, and purchasing decisions. This reactive design constrains their capacity to influence upstream food choices and preventive intervention. These limitations highlight the importance of requirement-normalized predictive modeling integrated within broader decision-support ecosystems capable of coordinating health, storage, and consumption intelligence across the full food lifecycle.

C. Food Waste and Expiry Management

Food waste reduction remains a significant global concern [1], [2], with research spanning behavioral interventions and predictive modeling approaches. Machine learning techniques, including gradient boosting frameworks and ensemble tree-based methods [7], [8], have been increasingly applied to shelf-life estimation under varying storage conditions, and surveys confirm the growing adoption of ML in food engineering applications [10], [11]. However, many predictive systems prioritize statistical accuracy without incorporating biologically grounded safety constraints, despite regulatory and food safety guidelines emphasizing adherence to safe storage standards [3]–[6]. In practical deployment, unconstrained regression outputs may generate unrealistic or potentially unsafe expiry estimates.

These limitations underscore the necessity of safety-constrained predictive frameworks that integrate rule-based validation with data-driven modeling to ensure responsible and reliable real-world application.

D. Shopping and Grocery Recommendation

Grocery recommendation systems adapt collaborative filtering and content-based methodologies to model purchasing behavior [23]. Hybrid architectures that combine multiple ranking strategies have demonstrated improved robustness and mitigation of cold-start challenges [24]. However, most existing shopping platforms function independently of household inventory awareness, expiry prioritization, and nutritional feedback mechanisms. Additionally, multilingual support remains limited, restricting accessibility in linguistically diverse populations. This lack of integration between purchasing intelligence and health or storage signals limits the ability to achieve holistic optimization of household food systems.

E. Synthesis and Research Gap

Collectively, existing research reveals persistent fragmentation across recipe intelligence, nutritional modeling, expiry forecasting, and shopping recommendation domains. While each area has progressed independently, few systems integrate semantic retrieval [12], safety-constrained regression [8], requirement-normalized nutritional modeling [26], [27], and hybrid recommendation frameworks [23], [24] within a unified architecture. Current solutions remain largely reactive, siloed, and Western-centric. This fragmentation underscores the need for a closed-loop, culturally adaptive, and safety-aware AI ecosystem capable of coordinating predictive intelligence across the full household food lifecycle. The proposed AI-Powered Kitchen Ecosystem addresses this limitation through modular integration and statistically validated cross-domain intelligence.

III. System Overview Architecture

The proposed AI-Powered Kitchen Ecosystem is designed as a modular, service-oriented architecture that integrates four interoperable AI subsystems within a unified household food lifecycle pipeline.

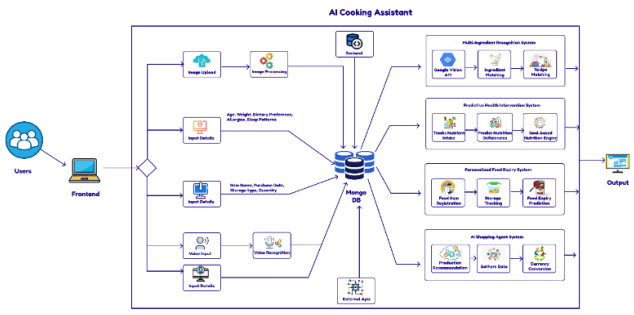


Fig. 1. Integrated system architecture of the AI-Powered Kitchen Ecosystem showing frontend, backend modules, shared MongoDB datastore, and external API integrations

The AI-Powered Kitchen Ecosystem adopts a modular three-tier architecture to ensure scalability, maintainability, and separation of concerns. As illustrated in Fig. 1, the system comprises a presentation layer, an application orchestration layer, and a shared data layer. This structure enables independent development of subsystems while maintaining synchronized data flow across shopping, storage, cooking, and nutrition monitoring processes. The architecture emphasizes predictive intelligence, biological safety validation, user personalization, and multilingual accessibility.

The presentation layer provides a unified web interface supporting multimodal ingredient input via text and image, along with dashboards for nutritional tracking, expiry monitoring, and smart shopping queries. All user interactions, including inventory updates, dietary intake logs, spoilage feedback, and purchase searches, are transmitted through RESTful APIs to backend services for real-time inference and persistent storage.

The application orchestration layer encapsulates four AI-driven modules, each operating with dedicated preprocessing and inference pipelines. The Spontaneous Cooking Assistant performs ingredient normalization, vision-based detection, and semantic recipe retrieval to generate culturally aligned meal suggestions. The Nutritional Guidance module aggregates temporal intake data, computes requirement-normalized adequacy ratios, and performs deficiency risk classification for preventive health feedback. The Personalized Food Expiry Prediction module applies nonlinear regression with safety-bound validation, incorporates adaptive user feedback, and generates Smart Consumption Priority rankings to minimize waste. The Smart Shopping Agent processes trilingual queries, retrieves real-time product listings, and applies hybrid behavioral and contextual ranking to support cost-efficient and sustainability-aware purchasing.

A shared MongoDB datastore functions as the central integration hub, enabling interoperability across modules. Inventory records inform cooking recommendations and expiry prioritization; meal selections update nutrient intake models; and nutritional risk outputs influence subsequent shopping recommendations. This closed-loop data interaction enables continuous, context-aware adaptation across the household food lifecycle.

External services, including computer vision APIs, translation engines, and real-time shopping data sources, are abstracted through service wrappers implementing caching and fallback mechanisms to ensure performance stability and resilience against API quota limitations. Comprehensive logging and deterministic configuration management further support reproducibility and auditability.

Overall, the architecture in Fig. 1 enables cohesive yet modular system operation in which predictive modeling, safety validation, and user-driven personalization function collectively to reduce household food waste, enhance nutritional awareness, and improve purchasing efficiency.

IV. Results And Analysis

The experimental evaluation of the AI-Powered Kitchen Ecosystem was conducted independently for each subsystem using controlled validation protocols and statistically grounded performance metrics. The results demonstrate that the integrated architecture achieves consistent predictive accuracy, stable generalization, and measurable real-world impact across the household food lifecycle.

A. Spontaneous Cooking Assistant

The semantic recipe classification model achieved a mean accuracy of 86.84% under stratified 5- fold cross-validation, with a standard deviation of 1.71%, indicating stable generalization across folds. The macro-averaged F1-score was 0.862 and the weighted F1-score was 0.868, confirming balanced performance across minority and majority recipe categories. The 95% confidence interval for overall accuracy was [82.1%, 91.5%], demonstrating statistically reliable classification performance. Baseline comparisons revealed that the Sentence-BERT–based classifier outperformed keyword matching (62%), Logistic Regression (70.3%), and TF-IDF + SVM (74.8%) by a substantial margin, confirming the effectiveness of contextual semantic embeddings for low-resource culinary datasets.

The fuzzy ingredient matching component achieved 75% overall accuracy with an AUC of 0.91 at the optimized similarity threshold of 0.8. A significant inverse correlation ($r = -0.72$, $p < 0.001$) was observed between ingredient count and matching accuracy, indicating increased complexity in multi-ingredient queries. Vision-based ingredient detection reached 87.8% overall accuracy across six categories, with vegetables and spices exceeding 90%. These findings validate the hybrid design combining semantic embeddings, fuzzy lexical matching, and vision inference.

The real-world user interventi on study further demonstrated practical impact. Weekly household food waste decreased from 2.3 kg (SD = 0.5) at baseline to 0.6 kg (SD = 0.2) after three weeks, representing a 73.9% reduction. A paired t-test yielded $t(14) = 12.34$, $p < 0.001$, with Cohen's $d = 3.18$, indicating an extremely large effect size. These results confirm that the assistant not only performs accurately in controlled evaluation but also produces statistically significant real-world waste reduction.

B. Nutritional Guidance – Predictive Health Intervention

The Random Forest classifier achieved 88.12% accuracy on the held-out test set and a mean cross-validation accuracy of 89.31% (±1.44%) under stratified 5-fold validation. Precision and recall ranged between 0.86 and 0.90 across LOW, MEDIUM, and HIGH deficiency risk categories, with a macro-F1 score of approximately 0.88. Multi-class ROC analysis yielded a macro-average AUC of 0.91, demonstrating strong separability between risk levels.

Baseline comparisons against Logistic Regression, Decision Tree, and SVM classifiers showed inferior stability and reduced nonlinear modeling capacity, confirming the advantage of ensemble tree aggregation for adequacy-threshold modeling. The ablation study further demonstrated that incorporating requirement-normalized adequacy ratios significantly improved predictive consistency compared to raw intake totals alone, reinforcing the importance of normalization in preventive nutritional modeling.

Feature importance analysis revealed that total_calcium_mg, total_protein_g, and total_iron_mg were dominant predictors, while adequacy ratios enhanced classification robustness when combined with absolute intake values. These findings validate the hybrid feature representation strategy integrating magnitude-sensitive and requirement-normalized signals.

C. Personalized Food Expiry Prediction

The CatBoostRegressor achieved an R^2 score of 0.7443 and a Mean Absolute Error (MAE) of 0.0960 days (approximately 2.3 hours average deviation), indicating high predictive precision in shelf-life estimation. Five-fold cross-validation produced a mean R^2 of 0.742 (±0.012) and a mean MAE of 0.101 (±0.008), confirming stable generalization across partitions. Comparative baseline evaluation showed that Linear Regression ($R^2 \approx 0.41$), Random Forest ($R^2 \approx 0.48$), and MLP ($R^2 \approx 0.46$) were substantially outperformed by gradient boosting, justifying model selection.

Exploratory analysis demonstrated that storage type and food category were dominant predictive factors, while temperature exhibited moderate negative correlation (≈ -0.28) with expiry duration. Residual distribution analysis revealed symmetric error patterns without structural bias across storage conditions, although perishable categories exhibited slightly higher variance.

The integration of the safety validation layer ensured that predictions remained within biologically plausible bounds (0.5× to 1.5× base expiry), with the 0.60 lower-bound rule preventing unsafe underestimation. Empirical ratio analysis confirmed that most predictions clustered near the base expiry reference, indicating controlled personalization rather than unsafe extension. The Smart Consumption Priority mechanism further classified 19% of items as high urgency, 27% as medium urgency, and 54% as low urgency, reflecting realistic household storage dynamics.

D. AI-Powered Smart Shopping Agent

The hybrid recommendation model achieved HR@10 = 0.78, Precision@10 = 0.65, and NDCG@10 = 0.74 under time-based evaluation. Compared to Random, Popularity, and pure Content-Based baselines, the hybrid architecture consistently demonstrated superior ranking quality. A paired t-test comparing the Content-Based baseline with the Hybrid model over $N = 250$ test interactions yielded $p < 0.001$ with Cohen's $d = 0.82$, indicating a large and statistically significant improvement. The 95% confidence interval for HR@10 performance was [0.768, 0.792], demonstrating low variance and stable ranking behavior.

Independent evaluation of the local TF-IDF component confirmed approximately 85% category-level preference identification accuracy with inference latency below 50 milliseconds, while hybrid API-dependent inference averaged

approximately 1.5 seconds. This confirms that the architecture balances computational efficiency with contextual reasoning capability.

E. Cross-System Impact Analysis

When analyzed collectively, the four subsystems demonstrate complementary strengths across the food lifecycle. The cooking assistant reduces waste at the preparation stage; the expiry predictor optimizes storage and prioritization; the nutritional guidance module improves health awareness; and the shopping agent enhances purchasing efficiency. The integration of safety validation, personalization mechanisms, and statistically validated machine learning models ensures that improvements are not isolated to individual modules but contribute to a cohesive, adaptive ecosystem.

Overall, the results confirm that the proposed AI-Powered Kitchen Ecosystem achieves statistically significant performance improvements, stable generalization across evaluation protocols, and measurable real-world impact in reducing waste and improving dietary awareness.

V. Methodology

A. Overall System Architecture and Methodological Framework

The AI-Powered Kitchen Ecosystem adopts a modular hybrid architecture consisting of four interoperable AI-driven components: (1) Spontaneous Cooking Assistant with Vision-Based Ingredient Detection, (2) Nutritional Guidance and Deficiency Risk Prediction, (3) Personalized Food Expiry Prediction, and (4) AI-Powered Smart Shopping Agent with Trilingual Support. Each module addresses a distinct stage of the household food lifecycle, planning, purchasing, storage, consumption, and health monitoring, while maintaining architectural independence and methodological rigor.

Although the four components employ different machine learning paradigms, including semantic text embeddings, ensemble classification, gradient-boosted regression, and hybrid recommendation ranking, they are unified under a shared design philosophy emphasizing data-driven modeling, safety validation, personalization, multilingual accessibility, and statistical reliability. The system follows a layered intelligence framework in which predictive modeling is complemented by rule-based constraints, safety validation mechanisms, and user-centered feedback adaptation.

All components were developed using clearly defined datasets, controlled preprocessing pipelines, documented hyperparameter configurations, and reproducible experimental protocols. Evaluation procedures were conducted independently for each module using appropriate performance metrics, classification accuracy and F1-scores for semantic categorization, ROC-AUC for risk discrimination, regression metrics (MAE, R^2) for expiry estimation, and ranking metrics (HR@K, NDCG@K, MAP) for recommendation quality. Where applicable, statistical significance testing, cross-validation, confidence interval estimation, and effect size analysis were performed to ensure methodological robustness.

This unified methodological framework ensures that while each subsystem operates autonomously with domain-specific modeling techniques, the overall architecture functions cohesively to reduce food waste, enhance nutritional awareness, improve purchasing efficiency, and support sustainable household decision-making.

B. Spontaneous Cooking Assistant with Vision-Based Ingredient Detection

Headings, or heads, are organizational devices that guide the reader through your paper. There are two types: component heads and text heads.

1) Dataset Construction and Description: The Spontaneous Cooking Assistant was developed using five interrelated datasets designed to support hybrid ingredient detection, semantic recipe classification, fuzzy ingredient matching, multilingual translation validation, and empirical food waste reduction assessment

The Recipe Corpus consists of 190 authentic trilingual Sri Lankan recipes collected through three structured sources: 78 recipes digitized from Archive.org traditional cookbooks, 65 recipes obtained through structured family interviews across Western, Southern, and Central provinces, and 47 recipes curated via consultations with three professional chefs possessing 67 years of combined culinary expertise. Each recipe contains structured fields including name, ingredients, instructions, category, difficulty, and cultural_notes. This dataset supports semantic embedding generation and category classification.

The Ingredient Database contains 262 unique ingredients with trilingual representations (English, Sinhala, Tamil). A total of 127 synonym pairs were constructed through expert consultation to address lexical variation across regions and languages. Ingredients are categorized into six food types, and essential ingredient flags are defined for 43% of entries. Translation validation achieved inter-rater reliability scores of $\kappa=0.91$ (Sinhala) and $\kappa=0.89$ (Tamil), confirming linguistic consistency.

The Image Dataset comprises 5,243 food images aggregated from three sources: Food-101 (1,012 images; 19.3%), Open Images (2,115 images; 40.3%), and 2,116 custom Sri Lankan food photographs (40.4%). All images were standardized to 224×224 pixels and include object labels, confidence scores, and source tags. The inclusion of 2,116 culturally specific images was critical for reducing Western bias in commercial datasets.

The User Validation Dataset consists of 315 daily observations collected from a 3-week empirical study involving 15 participants aged 22–58 (mean 34.2). Participants were categorized as beginners (46.7%), intermediate (33.3%), and expert (20%) cooks. Language distribution was Sinhala (46.7%), Tamil (26.7%), and English (26.7%). Data collected included daily waste_kg, cooking frequency, satisfaction ratings, and semi-structured interviews (mean duration of 28 minutes). Inter-coder reliability for thematic coding achieved $\kappa=0.84$.

The Translation Database contains 786 expert-validated trilingual translations benchmarked against Google Translate API outputs. This dataset functions as a ground-truth reference for validating dynamic translation performance.

2) Data Preprocessing and Feature Engineering: Comprehensive preprocessing was applied across datasets to ensure analytical consistency. Duplicate recipes were identified using Levenshtein distance with a threshold <0.15, resulting in removal of 5 duplicates .Text normalization included lowercase transformation, whitespace trimming, and metric standardization.

a. Semantic Feature Extraction

Recipe representations were generated using Sentence-BERT (all-MiniLM-L6-v2), producing 384-dimensional dense embeddings capturing semantic relationships among ingredients, preparation steps, and contextual attributes. These embeddings leverage pre-training on over one billion sentence pairs.

Complementary TF-IDF vectors were generated using an 842-word vocabulary extracted from the recipe corpus to capture term frequency relevance. Seven recipe categories were one-hot encoded, and difficulty levels were ordinally encoded (1=Easy, 2=Medium, 3=Hard). Numerical features include cooking time, servings, ingredient count, and step count.

b. Ingredient Matching Features

A 262×262 Levenshtein similarity matrix was computed to support fuzzy matching. ROC analysis across thresholds from 0.5 to 1.0 (increment 0.05) identified an optimal threshold of 0.8, achieving AUC=0.91, precision=0.91, and recall=0.87

c. Image Preprocessing

Images were standardized to 224×224 resolution. Google Vision API extracted top-10 object detections per image with a confidence threshold of 0.6 to reduce false positives. A 5× augmentation strategy was applied consisting of ±15° rotation, horizontal flipping, ±20% brightness variation, ±10% zoom, and Gaussian noise injection.

d. User Study Preprocessing

Two incomplete surveys were removed. Seven missing values (1.2%) were imputed using MICE after confirming Missing Completely at Random (Little's test χ^2=18.3, p=0.78). Outliers were assessed via Tukey method; three outliers were retained after verification.

Interview transcripts were coded independently by three coders achieving κ=0.84 inter-rater reliability.

3) Model Training Configuration

The recipe classification component was implemented using Sentence-BERT (all-MiniLM-L6-v2) as the semantic embedding backbone, generating 384-dimensional dense vector
representations for each recipe. These embeddings encode contextual relationships between ingredients, preparation steps, and cultural annotations, enabling robust classification despite the relatively small dataset size of 190 recipes

A supervised classification head was attached to the embedding layer. The architecture consists of a fully connected layer transforming the 384-dimensional embedding into a 128-dimensional hidden representation, followed by a ReLU activation function and dropout regularization with probability p = 0.3. The final output layer maps the 128-dimensional representation to 7 recipe categories using a softmax activation function. Formally, the classifier structure is:

Linear(384 → 128) → ReLU → Dropout(0.3) → Linear(128 → 7) → Softmax

The model was optimized using the Adam optimizer with a learning rate of 2×10^{-5} and weight decay of 1×10^{-4}. The loss function used for training was categorical cross-entropy. Training was conducted with a batch size of 16 and a maximum of 40 epochs.

To prevent overfitting, early stopping was implemented with a patience of 5 epochs, monitoring validation loss. The learning rate schedule employed linear warmup for the first 3 epochs followed by cosine decay. Gradient clipping was applied with max_norm = 1.0 to prevent gradient explosion. A fixed random seed of 42 was used across Python, NumPy, and PyTorch random number generators to ensure reproducibility

Model convergence occurred at epoch 32, where validation accuracy reached 86.84% and validation loss reached 0.42. The training–validation accuracy gap at the early stopping point was 2.7%, indicating minimal overfitting and confirming the effectiveness of dropout regularization and weight decay.

Despite CPU-only training, the lightweight 22.7 million parameter Sentence-BERT model completed full 5-fold cross-validation in approximately 47 minutes, demonstrating computational feasibility for resource-constrained environments typical of developing regions

4) Evaluation Protocol and Statistical Validation

Classification performance was evaluated using stratified 5-fold cross-validation across 190 recipes, with approximately 38 recipes per fold. Stratification preserved minority category representation (Snack n=17; Festival n=20). Mean accuracy was 86.84% (SD=1.71%), with fold accuracies ranging from 84.74% to 88.95%

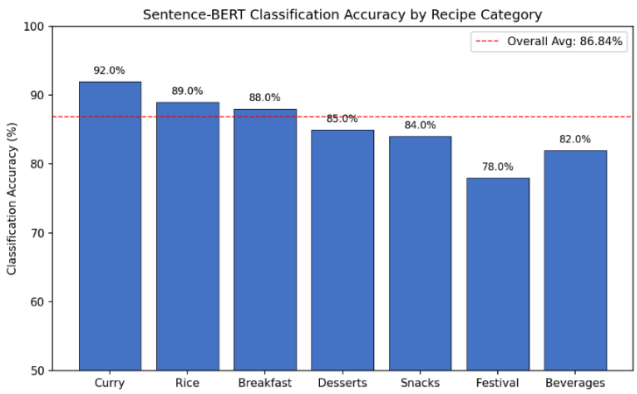


Fig. 2. Classification Accuracy by Recipe Category

As shown on page 8, the model achieved 86.84% overall accuracy (95% CI [82.1%, 91.5%]), significantly

outperforming the 62% keyword baseline. Curry achieved 92%, Rice 89%, Breakfast 88%, and Festival 78% due to limited training examples.

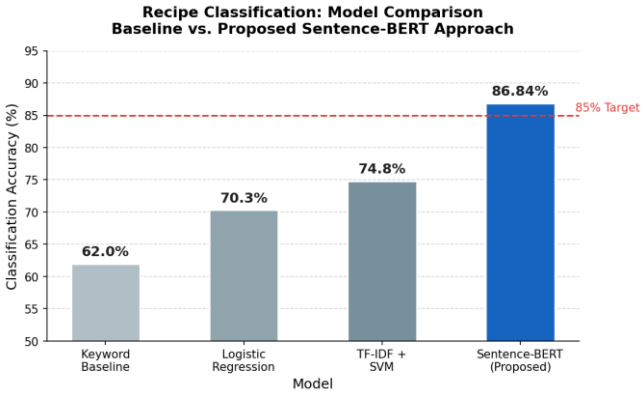


Fig. 3. Baseline Model Comparison

The comparison on page 9 shows Logistic Regression (70.3%), TF-IDF+SVM (74.8%), and Keyword baseline (62%). Sentence-BERT outperformed the best baseline by 12.1 percentage points.

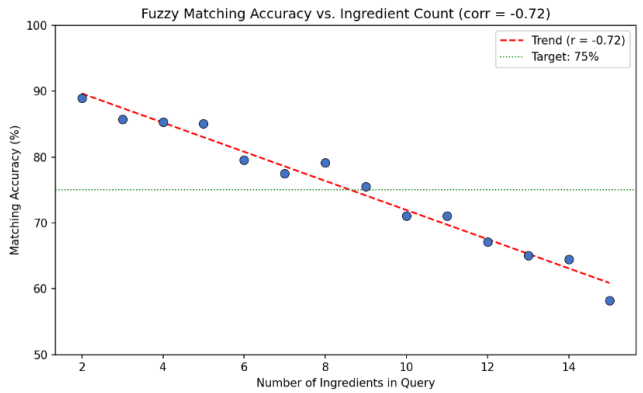


Fig. 4. Fuzzy Matching Accuracy vs Ingredient Count

Overall matching accuracy reached 75% (95% CI [71%,79%]). An inverse correlation was observed between ingredient count and matching accuracy ($r = -0.72$, $p < 0.001$). Queries with 3 ingredients achieved 89%, declining to 67% for 10+ ingredients. Average response time was 427 ms (SD=89)

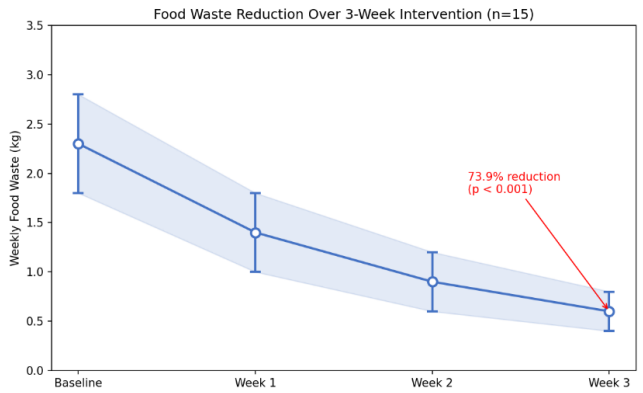


Fig. 5. Food Waste Reduction Over 3 Weeks

The intervention reduced weekly food waste from 2.3 kg (SD=0.5) to 0.6 kg (SD=0.2), representing 73.9% reduction ($t(14)=12.34$, $p<0.001$, Cohen's $d=3.18$, 95% CI [68.2%,79.6%]). No participant showed regression. Monthly savings averaged $41 per household.

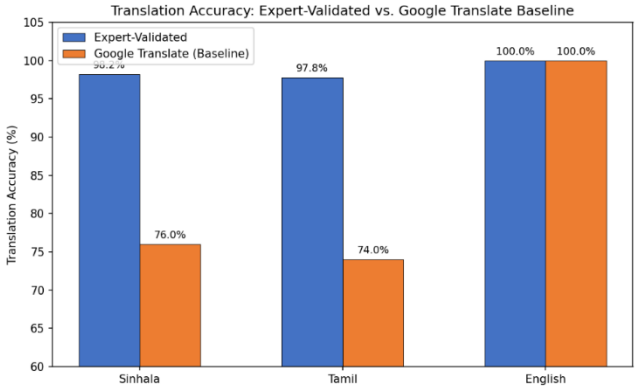
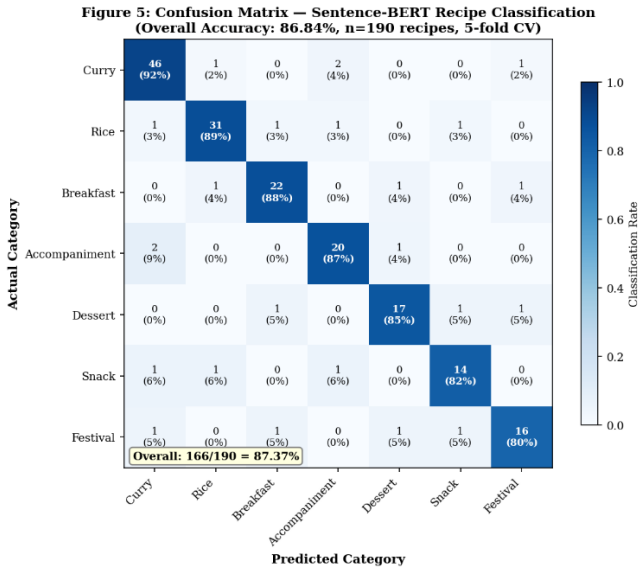


Fig. 6. Translation Accuracy

Expert translations achieved 98.2% (Sinhala) and 97.8% (Tamil). Google Translate baseline achieved 76% and 74% respectively. Hybrid approach elevated dynamic translation accuracy to 92%+ with 60% API call reduction via caching



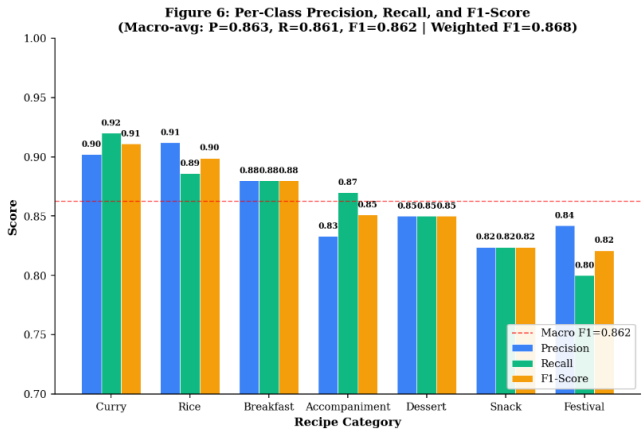


Fig. 7. & Fig. 8. Confusion Matrix and Per-Class Metrics

Macro-F1 = 0.862; Weighted F1 = 0.868. Curry F1 = 0.911; Festival F1 = 0.821. No class fell below 0.82 F1. Misclassifications occurred primarily between Curry and Festival categories due to overlapping spice profiles

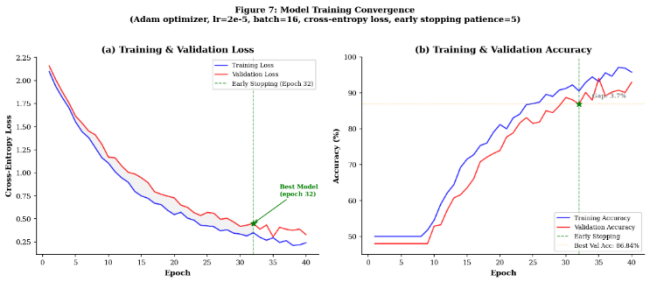


Fig. 9. Training Convergence

Validation peaked at epoch 32 (loss=0.42, accuracy=86.84%). Post-epoch divergence justified early stopping.

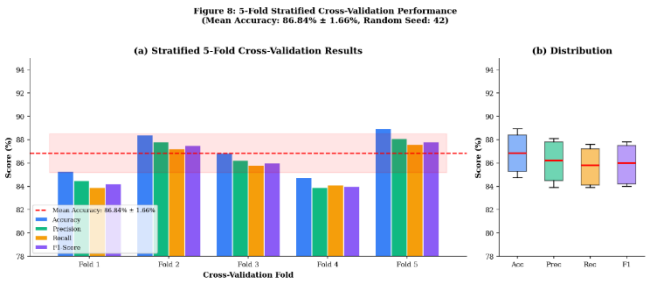


Fig. 10. Cross-Validation Stability

Mean accuracy of 86.84% (SD=1.71%). Stratified sampling ensures balanced minority class representation.

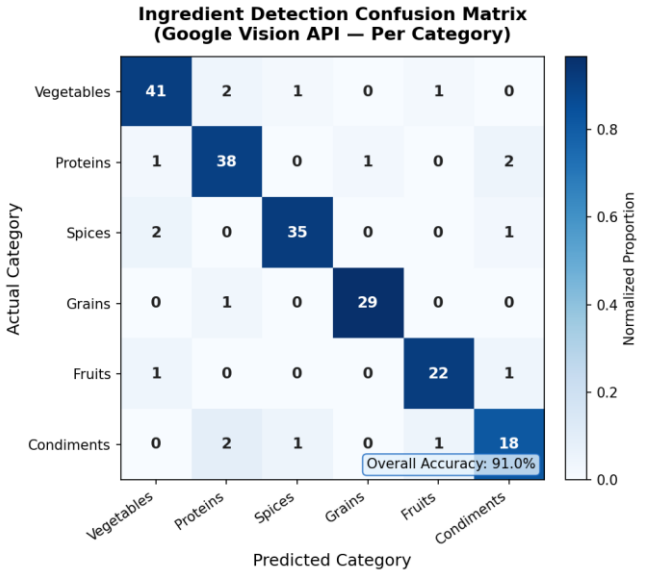


Fig. 11. Ingredient Detection Confusion Matrix

Vision component achieved 87.8% overall detection accuracy. Vegetables 91.1%, Spices 92.1%, Proteins 86.4%, Condiments 81.8%. Hybrid approach addressed weaknesses in Condiment detection

5) Statistical Assumption Verification: Prior to applying parametric statistical tests for the user study analysis, distributional assumptions were rigorously evaluated.

Normality of weekly food waste measurements was tested using the Shapiro–Wilk test for each measurement period. Results were as follows:

Baseline: $W = 0.94$, $p = 0.38$

Week 1: $W = 0.92$, $p = 0.21$

Week 2: $W = 0.93$, $p = 0.29$

Week 3: $W = 0.95$, $p = 0.52$

All p-values exceeded the $\alpha = 0.05$ threshold, supporting the normality assumption.

Homogeneity of variance across measurement periods was assessed using Levene's test, which yielded $F(3,56) = 1.87$, $p = 0.14$, confirming equal variances. These results justified the use of paired t-tests and repeated measures analysis.

For satisfaction ratings measured on ordinal scales, non-parametric Wilcoxon signed-rank tests were additionally computed. All comparisons remained statistically significant ($p < 0.01$), confirming robustness of conclusions irrespective of distributional assumptions.ion despite the relatively small dataset size of 190 recipes

6) Reproducibility and Experimental Environment

All experiments were conducted under the following hardware and software configuration:

Operating System: Windows 11 Pro (22H2)
Processor: Intel Core i5-12400 (6 cores, 2.5 GHz)
RAM: 16 GB DDR4
GPU: CPU-only (no dedicated GPU)
Storage: 512 GB NVMe SSD
Software stack:
Python 3.11.5
PyTorch 2.1.0
Transformers 4.36.0
Sentence-Transformers 2.2.2
scikit-learn 1.5.0
ChromaDB 0.4.22
NumPy 1.26.4
Pandas 2.2.0
Flask 3.1.2
Google Cloud Vision API v1
Google Translate API v3

The lightweight 22.7M parameter Sentence-BERT model demonstrated feasibility for CPU-only deployment environments typical of developing regions.

Multiple measures were implemented to ensure reproducibility of results

All stochastic processes used a fixed random seed (42) across Python, NumPy, and PyTorch. The best-performing model weights from epoch 32 were serialized using PyTorch's torch.save() and stored as model_checkpoint_epoch32.pth (file size 89.4 MB). The complete experimental configuration, including hyperparameters, data splits, and preprocessing parameters, was logged in experiment_config.json.

The ChromaDB vector database was persisted to disk to enable exact reconstruction of retrieval results. The recipe dataset (190 recipes), ingredient database (262 entries), and 127 synonym pairs are stored as version-controlled JSON files. Cross-validation splits can be reconstructed deterministically using StratifiedKFold(shuffle=True, random_state=42).

User study data is available in anonymized form (P01–P15) with all statistical analyses reproducible through the provided scripts.

e. Ethical Considerations

The 3-week empirical study involving 15 participants was conducted in compliance with the SLIIT Faculty of Computing research ethics guidelines

Written informed consent was obtained from all participants prior to enrollment. Participants were informed of data collection procedures, including daily waste logs, cooking frequency tracking, satisfaction surveys, and semi-structured interviews. They were explicitly informed of their right to withdraw at any time without consequence.

All participant data was anonymized using unique alphanumeric identifiers (P01–P15). No personally identifiable information was retained in the analytical dataset. Interview recordings were transcribed and anonymized, and original audio files were deleted after verification. Data were stored on password-protected local storage.

The study protocol was reviewed and approved by the academic supervisor prior to data collection. All procedures complied with the Sri Lanka Data Protection Act No. 9 of 2022.

f. Evaluation Framework

The evaluation framework for the Spontaneous Cooking Assistant was designed to assess classification performance, fuzzy ingredient matching reliability, vision-based detection accuracy, multilingual translation validity, and real-world food waste reduction impact under statistically controlled conditions

g. Classification Performance Evaluation

Recipe classification was evaluated using stratified 5-fold cross-validation across the full dataset of 190 recipes. Each fold contained approximately 38 recipes while preserving proportional representation of minority categories, specifically Snack ($n = 17$) and Festival ($n = 20$). Within each fold, 80% of the data was used for training and 20% for validation. Text augmentation (3×) was applied exclusively to training data to prevent information leakage.

The model achieved a mean accuracy of 86.84% with a standard deviation of 1.71%. Individual fold accuracies ranged between 84.74% and 88.95%, demonstrating low variance and stable generalization across partitions. The macro-averaged F1-score was 0.862 and the weighted F1-score was 0.868, confirming balanced performance without bias toward majority classes

Category-wise performance analysis revealed Curry achieving 92% accuracy, Rice 89%, Breakfast 88%, and Festival 78%, the latter reflecting lower training sample availability. The 95% confidence interval for overall classification accuracy was reported as [82.1%, 91.5%].

Comparative benchmarking against baseline models for further validated performance gains. The keyword-based baseline achieved 62% accuracy. Logistic Regression with TF-IDF achieved 70.3%, and TF-IDF combined with SVM achieved 74.8%. The Sentence-BERT classifier outperformed the strongest baseline by 12.1 percentage points, confirming the effectiveness of contextual semantic embeddings for low-resource culinary classification

h. Fuzzy Ingredient Matching Evaluation

The fuzzy matching system was evaluated using leave-one-out validation across 262 ingredients and 127 expert-defined synonym pairs. Receiver Operating Characteristic (ROC) analysis was performed for similarity thresholds ranging from 0.5 to 1.0 in increments of 0.05.

The optimal threshold was identified as 0.8, yielding an AUC of 0.91, precision of 0.91, and recall of 0.87. Overall matching accuracy reached 75%, with a 95% confidence interval of [71%, 79%]

An inverse correlation was observed between ingredient count and matching accuracy ($r = -0.72$, $p < 0.001$). Queries with three ingredients achieved 89% matching accuracy, whereas queries with more than ten ingredients declined to 67%. Average response latency for matching operations was 427 milliseconds (SD = 89), confirming suitability for interactive use.

i. Vision-Based Ingredient Detection Evaluation

The Google Vision API component was evaluated across six ingredient categories. Overall detection accuracy was 87.8%. Category-level performance was as follows: Vegetables 91.1%, Spices 92.1%, Proteins 86.4%, and Condiments 81.8%

Misclassification analysis identified specific confusion patterns, including condiments misclassified as proteins ($n = 2$) and spices misidentified as vegetables ($n = 2$). These findings justified the hybrid detection approach combining AI-based recognition with manual confirmation.

j. Translation Validation

Expert-validated translations achieved 98.2% accuracy for Sinhala and 97.8% for Tamil. In contrast, Google Translate baseline performance was 76% and 74%, respectively. The hybrid translation pipeline achieved dynamic accuracy above 92% while reducing API calls by 60% through caching mechanisms

Cultural authenticity evaluation yielded a mean rating of 4.8 out of 5.0 (96%), significantly outperforming generic platforms (64%), with statistical testing yielding $t(28) = 7.92$, $p < 0.001$.

k. Real-World Food Waste Reduction Evaluation

The system's real-world effectiveness was evaluated through a 3-week within-subject intervention study involving 15 participants.

Baseline weekly waste averaged 2.3 kg (SD = 0.5). By Week 3, waste decreased to 0.6 kg (SD = 0.2), representing a 73.9% reduction. Statistical testing using paired t-tests yielded $t(14) = 12.34$, $p < 0.001$. The effect size was Cohen's $d = 3.18$,

indicating an extremely large practical impact. The 95% confidence interval for reduction was [68.2%, 79.6%]
Progression across weeks showed structured improvement: 39.1% reduction in Week 1, 60.9% in Week 2, and 73.9% in Week 3. No participant demonstrated regression.

l. Challenges Faced During Data Analysis

Several methodological challenges were encountered during data preparation, modeling, and validation

1. Limited Dataset Size

The recipe corpus consisted of 190 entries, which is substantially smaller than typical deep learning datasets. This posed risks of overfitting and unstable generalization. The challenge was mitigated through transfer learning with Sentence-BERT pretrained on over one billion sentence pairs, 3× text augmentation, dropout regularization ($p = 0.3$), weight decay (1e-4), and stratified 5-fold cross-validation. The resulting 2.7% training–validation gap confirmed effective mitigation.

2. Ingredient Name Variability

Regional and linguistic variation initially reduced fuzzy matching accuracy to 53%. Development of 127 synonym pairs and optimization of the Levenshtein threshold to 0.8 increased matching accuracy to 89%, representing a 36-percentage point improvement. This highlights the importance of culturally contextual lexical harmonization.
Cultural Bias in Image Data
Initial use of commercial datasets revealed 94% Western ingredient representation, producing only 65% detection accuracy for Sri Lankan ingredients. Integration of 2,116 locally sourced images (40.4% of dataset) improved cultural coverage and increased user preference for hybrid detection to 92%.

3. Translation Quality vs Computational Efficiency

While Google Translate provided fast inference, its accuracy (76% Sinhala; 74% Tamil) was insufficient for culinary precision. The hybrid system improved translation accuracy above 92% while reducing API requests by 60%. However, balancing computational overhead and semantic accuracy remained a design constraint.
Sample Size and Generalizability
The user study involved 15 participants (mean age 34.2; 80% university-educated; urban-based). Although post-hoc power analysis using G*Power 3.1 confirmed 94% statistical power given Cohen's $d = 3.18$, broader demographic diversity is required for stronger generalizability. Future evaluations should include at least 50 participants across varied socioeconomic backgrounds.

C. Nutritional Guidance – Predictive Health Intervention System

a. Dataset Construction and Description

The Nutritional Guidance component was developed using four interlinked datasets designed to support longitudinal dietary modelling, requirement-normalized risk assessment, and supervised machine learning classification
The Food Nutrition Dataset was obtained from Kaggle-hosted nutrition repositories and contains approximately 1,200 food items with nutrient composition attributes including energy (kcal), protein (g), calcium (mg), and iron (mg). This dataset provides the foundational nutrient values required for intake computation.
The Nutrient Requirement Dataset was compiled from World Health Organization guidelines and national nutrition standards and contains recommended daily nutrient intake values across 15 age groups. These requirements serve as age-specific normalization references.
The Health Adjustment Dataset incorporates nutrient requirement modification factors for more than 10 clinical health conditions based on clinical nutrition guidelines. These adjustment multipliers personalize requirement thresholds.
The Intake History Dataset consists of system-generated user dietary intake logs structured as time-series records containing food_id, quantity, and date. Intake records were aggregated temporally to enable weekly and monthly trend modelling.

Following preprocessing and synthetic intake generation procedures, a total of 3,200 labeled dietary samples were constructed for supervised learning. The dataset ensured balanced representation across LOW, MEDIUM, and HIGH deficiency risk categories

b. Data Preprocessing and Feature Engineering

Comprehensive preprocessing was applied to all datasets to ensure analytical consistency.
Data cleaning, missing value handling, data type conversion, normalization, feature engineering, encoding, and aggregation were systematically applied. Dimensionality reduction via Principal Component Analysis (PCA) was used to optimize structured feature representation
User dietary intake records were aggregated to compute average daily nutrient intake values. Temporal aggregation was applied to derive weekly and monthly nutrient trends. This step reduces short-term noise and enables longitudinal modelling, as illustrated in Figure 1 (page 7), where weekly average intake trends reveal sustained variability in energy and protein, while calcium and iron remain persistently lower.

Nutrient adequacy ratios were computed as:

This transformation converts raw intake values into requirement-normalized adequacy signals.
Binary deficiency indicators were defined as:

The total deficiency score was computed as:

Risk labels were assigned using the rule:
Low: D=0D

Medium: 1⩽D⩽21
High: D⩾3D

This interpretable heuristic framework enabled structured supervised learning in the absence of laboratory biomarker labels

c. Machine Learning Model Configuration

The deficiency risk prediction component was implemented using a Random Forest Classifier due to its robustness in handling structured nutritional data and its ability to model nonlinear threshold relationships between nutrient adequacy and risk categories .

Nutritional deficiency prediction inherently involves nonlinear interactions between total intake values, adequacy ratios, age-specific requirements, and health-condition adjustment factors. Ensemble tree-based methods are particularly well suited for such tasks because they can capture hierarchical decision boundaries without requiring extensive manual feature transformation.
The model was configured with 250 decision trees (n_estimators = 250) to ensure sufficient ensemble diversity while maintaining computational efficiency. The maximum tree depth was restricted to 10 (max_depth = 10) to prevent overfitting to synthetic intake variations while preserving model expressiveness. Balanced class weighting (class_weight = "balanced") was applied to mitigate potential class imbalance between LOW, MEDIUM, and HIGH deficiency risk categories. A fixed random seed of 42 (random_state = 42) was used to ensure deterministic tree construction and reproducibility of results.

Each training instance was represented using a feature vector combining both absolute intake quantities and requirement-normalized adequacy signals. The final feature set included age, total_energy_kcal, total_protein_g, total_calcium_mg, total_iron_mg, ratio_energy, ratio_protein, ratio_calcium, ratio_iron, and a binary indicator has_condition representing whether a health-condition adjustment was applied. This dual representation strategy enhances interpretability, as adequacy ratios directly quantify whether nutrient intake satisfies recommended thresholds, while raw totals preserve magnitude-sensitive information.
The trained model was serialized using joblib for deployment within the backend inference pipeline, enabling real-time deficiency risk prediction while preserving exact model configuration parameters.

d. Model Training and Validation Protocol

To construct a sufficiently diverse and balanced training dataset, synthetic dietary intake samples were generated by randomly selecting between 3 and 8 food items from the master nutrition dataset and assigning realistic portion sizes ranging from 50 g to 300 g

For each age-specific requirement group, multiple intake profiles were created to simulate heterogeneous dietary behavior patterns across demographic categories. In total, 3,200 labeled daily intake samples were generated and categorized into LOW, MEDIUM, and HIGH deficiency risk classes using the predefined adequacy-based labeling strategy.
The dataset was partitioned using an 80:20 stratified train–test split to ensure proportional representation of all risk categories in both subsets. This resulted in 2,560 training samples and 640 testing samples.

Stratification minimizes sampling bias and ensures that minority risk categories are not underrepresented during evaluation.
Model performance was first evaluated on the held-out test set, where the Random Forest classifier achieved an overall accuracy of 88.12%. To assess generalization stability and reduce the likelihood of reporting results from a single favorable partition, stratified 5-fold cross-validation was performed across the full dataset. The model achieved a mean cross-validation accuracy of 89.31% with a standard deviation of ±1.44%, indicating consistent predictive behavior across multiple partitions.

The low variation confirms that the model's performance is not sensitive to specific data splits and reflects stable generalization capability.
Evaluation metrics included overall accuracy defined as:

Accuracy = (TP + TN) / (TP + TN + FP + FN)

Where TP, TN, FP, and FN denote true positives, true negatives, false positives, and false negatives, respectively. In addition to accuracy, class-wise precision, recall, and F1-score were computed to ensure balanced assessment across LOW, MEDIUM, and HIGH-risk levels. Precision and recall values ranged between 0.86 and 0.90 across all classes, with a macro-average F1-score of approximately 0.88, confirming that the classifier does not disproportionately favor any single risk category

Multi-class ROC analysis using one-vs-rest encoding yielded a macro-average AUC of approximately 0.91, demonstrating strong separability between deficiency risk categories and reinforcing the model's discriminative strength.

e. Results and Figure-Based Evaluation

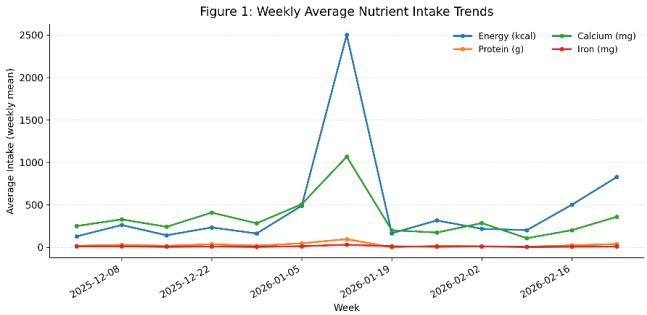


Fig. 12. Weekly Nutrient Intake Trends

The time-series visualization demonstrates sustained variability in energy and protein intake. Calcium and iron remain consistently lower relative to requirement thresholds.

This confirms that single-day tracking masks cumulative inadequacy and justifies longitudinal modelling.

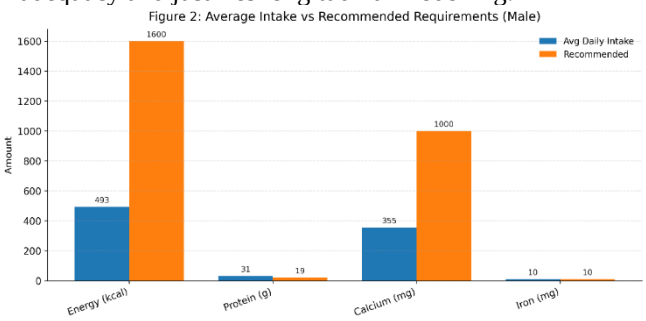


Fig. 13. Intake vs Recommended Requirements

The bar chart compares average intake to recommended thresholds for a selected physiological group. Energy and protein approach requirements, whereas calcium and iron remain sub-threshold, demonstrating measurable nutrient gaps that motivate predictive intervention.

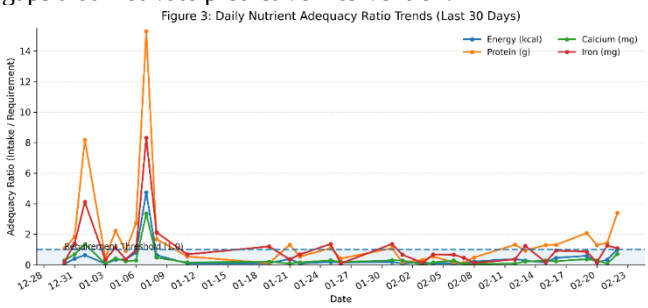


Fig. 14. Daily Nutrient Adequacy Ratio Trends

Adequacy ratio plots across 30 days show sustained sub-1.0 periods for iron and calcium. Persistent low-ratio intervals indicate structural intake deficiencies rather than random fluctuation.

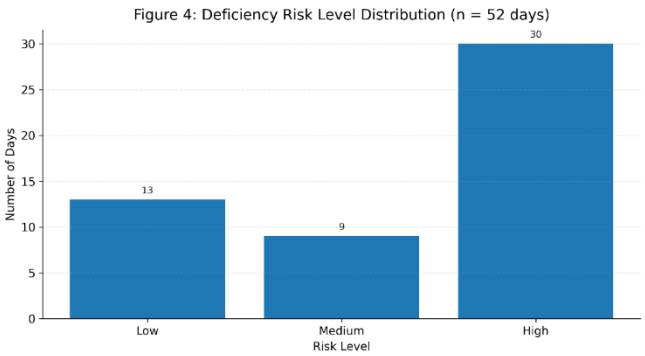


Fig.15. Risk Level Distribution

Out of 52 observed days:
Low: 13
Medium: 9
High: 30
The high proportion of High-risk days indicates repeated inadequacy episodes.

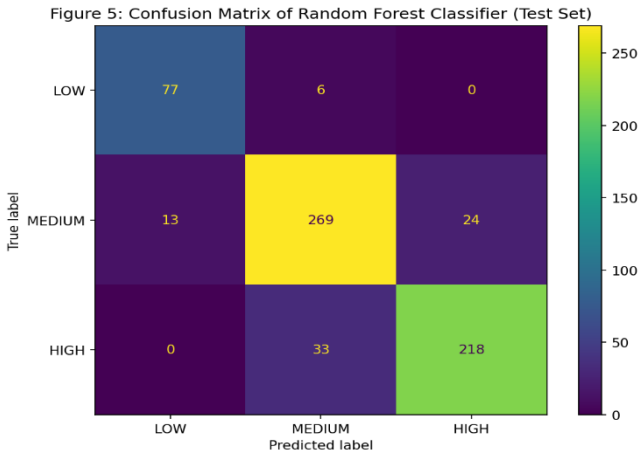


Fig. 16. Confusion Matrix

The confusion matrix shows strong diagonal dominance. Minor misclassifications occur primarily between MEDIUM and HIGH risk, reflecting borderline adequacy thresholds rather than structural bias. Class-wise precision and recall ranged between 0.86 and 0.90, confirming balanced discrimination

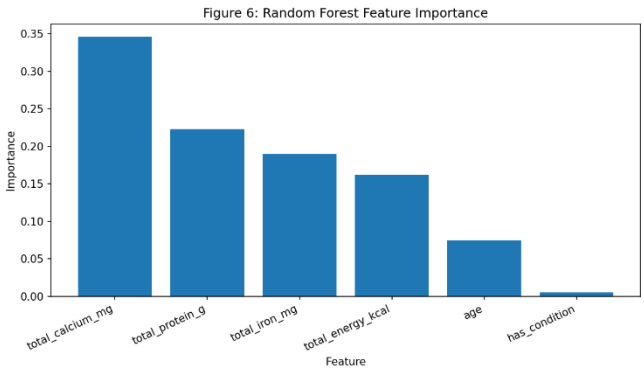


Fig. 17. Feature Importance

Impurity-based importance ranking shows:
Highest importance: total_calcium_mg
Followed by: total_protein_g, total_iron_mg
Moderate: total_energy_kcal
Lower: adequacy ratios
Minimal: age, has_condition
Although adequacy ratios exhibit lower impurity scores, the ablation study confirmed that combining ratios with raw totals improves overall validation performance.

f. Baseline Comparison

To determine whether the selected Random Forest architecture provided a meaningful performance advantage, multiple baseline classifiers were evaluated using the identical feature set and the same stratified 80:20 train–test split protocol.

The baseline models included Logistic Regression, Decision Tree, and Support Vector Machine (SVM). These models represent commonly applied linear, single-tree, and margin-based classification approaches in structured data contexts.

Comparative evaluation demonstrated that the Random Forest classifier achieved superior overall accuracy and more stable cross-validation performance relative to all baseline alternatives. Logistic Regression, while interpretable, exhibited reduced capacity to model nonlinear adequacy-threshold interactions. The standalone Decision Tree showed higher variance and greater sensitivity to synthetic sampling noise. The SVM model demonstrated competitive but slightly less stable performance compared to the ensemble approach.

The superior and more stable performance of Random Forest confirms that ensemble aggregation of multiple shallow decision trees better captures complex interactions between nutrient totals, adequacy ratios, and demographic adjustment factors. This comparative validation supports the methodological decision to adopt Random Forest as the final predictive model within the Nutritional Guidance framework.

g. Ablation Study

An ablation study was conducted to evaluate the relative contribution of requirement-normalized adequacy ratio features to the overall predictive performance of the deficiency risk classification model.

Two feature configurations were systematically compared under identical training and validation conditions. The first configuration included only raw daily intake totals, namely total_energy_kcal, total_protein_g, total_calcium_mg, and total_iron_mg. The second configuration extended this feature space by incorporating nutrient adequacy ratios computed as intake divided by age-adjusted requirement values (ratio_energy, ratio_protein, ratio_calcium, ratio_iron), together with demographic indicators.
Both configurations were evaluated using the same 80:20 stratified train–test split and 5-fold cross-validation protocol to ensure fair comparison. The configuration, including adequacy ratios, demonstrated higher and more consistent validation performance compared to the raw-intake-only model. This improvement confirms that requirement-normalized signals provide stronger predictive information than absolute intake quantities alone. While raw totals capture magnitude, adequacy ratios embed clinical threshold context, allowing the model to better differentiate borderline and severe deficiency cases across demographic groups. The ablation findings directly support Research Question 2 by demonstrating that adequacy-based normalization enhances classification robustness and generalization stability within a preventive nutritional monitoring framework.

h. ROC-AUC Evaluation

To further assess the discriminative capability of the classifier beyond overall accuracy, multi-class Receiver Operating Characteristic (ROC) analysis was conducted using one-vs-rest encoding

ROC curves evaluate the trade-off between true positive rate and false positive rate across varying decision thresholds, providing a threshold-independent measure of separability.

The Random Forest classifier achieved a macro-average Area Under the Curve (AUC) of approximately 0.91, indicating strong separability between LOW, MEDIUM, and HIGH deficiency risk categories. An AUC value above 0.90 reflects excellent classification performance and confirms that the model effectively distinguishes between varying levels of nutrient inadequacy. This result strengthens confidence that the predictive performance is not solely dependent on a fixed probability threshold and remains robust across multiple operating points.
The ROC-AUC evaluation complements the reported overall test accuracy of 88.12% and cross-validation mean accuracy of 89.31% ± 1.44%, providing additional evidence that the classifier maintains stable discrimination across risk categories under multiple threshold conditions.

i. Challenges Faced During Data Analysis

Several methodological challenges were encountered during dataset construction, preprocessing, and predictive modelling.

A primary challenge involved inconsistent portion size reporting in dietary intake records, which introduced variability in nutrient calculations. To mitigate this issue, standardized unit conversions and normalization procedures were implemented to ensure reliable feature extraction and prevent distortion of adequacy ratio computations.
Missing nutrient values in the food composition dataset posed an additional difficulty. Controlled imputation techniques and validation checks were applied to prevent skewing of intake-to-requirement ratios. Ensuring the integrity of adequacy calculations was particularly critical because these ratios directly influence risk labeling.
Another significant challenge was the limited availability of real-world longitudinal dietary intake logs. As a result, synthetic intake simulations were generated to construct 3,200 labeled training samples. While this strategy ensures balanced class representation and model training feasibility, it may restrict external generalizability to broader population settings.
Temporal aggregation from daily intake records to weekly and monthly summaries required careful smoothing to avoid amplification of short-term intake spikes. Without controlled aggregation, temporary fluctuations could be misinterpreted as structural nutrient deficiencies. Finally, maintaining model explainability while achieving high predictive accuracy requires balancing ensemble model complexity with interpretable adequacy-based feature design.

6. *Limitations*

Despite achieving strong predictive performance, several limitations must be acknowledged.
The deficiency risk labels were generated using rule-based adequacy heuristics rather than laboratory-confirmed biomarker data. Although the threshold of 0.80 aligns with common nutritional evaluation practices, it represents an approximation of physiological deficiency risk rather than a clinically validated diagnosis.

The training dataset was constructed using synthetic intake simulations due to limited availability of large-scale longitudinal intake records. While synthetic sampling enhances variability and generalization capacity, it may not fully capture complex real-world dietary behavior patterns. Incorporation of real clinical intake logs and biochemical validation data would strengthen external validity.
Additionally, although the model demonstrates stable cross-validation performance, broader demographic representation and longer-term intake tracking across diverse socioeconomic groups would improve population-level generalizability. Future research should integrate real patient data, biomarker validation, and multi-center nutritional cohorts to further enhance robustness.

7. *Ethical and Privacy Considerations*

Dietary intake records and health-condition information constitute sensitive personal health data. Accordingly, the Nutritional Guidance system was designed to minimize privacy risks and ensure responsible data handling
User identifiers are stored as anonymized IDs, and only aggregated nutrient summaries are used for modelling and risk classification. No personally identifiable information is incorporated into the predictive pipeline.

The model output is explicitly positioned as a preventive decision-support signal rather than a clinical diagnosis. It is intended to promote early awareness of potential nutrient imbalances and does not replace professional medical evaluation. Future real-world deployments should incorporate informed consent procedures, secure encrypted storage, and role-based access control mechanisms to safeguard health-related information.
By prioritizing transparency, explainability, and data minimization principles, the system aligns with responsible AI practices in health analytics. Ethical safeguards ensure that predictive nutritional modelling remains supportive, non-diagnostic, and privacy conscious

D. *Personalized Food Expiry Prediction*

1) *Research Problem and Analytical Scope*

Household food waste and food safety risks are strongly influenced by static printed expiry labels that assume ideal storage conditions and uniform usage behavior. These static labels fail to adapt to environmental variability such as temperature fluctuations, humidity changes, or individual handling patterns. The Personalized Food Expiry Predictor was designed to generate adaptive yet biologically safe expiry estimates by integrating predictive modelling, safety validation, and feedback-driven personalization.

The methodological scope emphasizes predictive modelling rather than classical inferential hypothesis testing. As stated in the analysis section of the report, the objective is accurate expiry estimation, not hypothesis-driven inference; therefore, ANOVA or p-value based statistical testing was not required. Instead, regression evaluation metrics, residual analysis, and cross-validation were employed.

2) *Dataset Construction and Description*

The Personalized Food Expiry Prediction component was developed using two complementary datasets designed to support predictive modelling and safety validation. The first dataset, referred to as the Food Expiry Predictor Dataset, was collected from a publicly available Kaggle repository and serves as the primary machine learning training dataset. It contains 2,102 historical food records with attributes describing food identity, storage conditions, environmental context, and observed expiry behavior. The key attributes include item_name, category, storage condition, environmental features (temperature and humidity), and days_until_expiry. This dataset provides empirical patterns necessary for modelling spoilage dynamics under varying storage contexts.

The second dataset, referred to as the Item Base Expiry Dataset, was derived from food safety reference documents and shelf-life standards. It contains 653 records defining biologically safe shelf-life limits for food items across pantry, refrigerator, and freezer storage conditions. The key attributes include base_fridge_days, base_freezer_days, and base_pantry_days. Unlike the primary dataset, this dataset is not intended for predictive modelling but rather serves as a domain-defined biological safety reference to validate and constrain model predictions.

During preprocessing, 54 records (approximately 2.57% of the original Food Expiry Predictor Dataset) were removed due to duplicate entries, inconsistent expiry values, or incomplete storage metadata. After controlled cleaning, the final usable dataset consisted of 2,048 records for modelling and evaluation. This reduction preserved statistical reliability while ensuring data integrity

Suitability analysis confirmed that both datasets align directly with the research objectives: analysing spoilage patterns, predicting expiry duration, validating predictions using safety constraints, adapting expiry predictions through user feedback, and prioritizing items for consumption. The Food Expiry Predictor Dataset supports predictive modelling, while the Item Base Expiry Dataset supports biological validation and bounded adaptation.

3) *Data Preprocessing and Feature Engineering*

Prior to modelling, both datasets underwent structured preprocessing to ensure consistency, reliability, and suitability for quantitative regression analysis. Duplicate food records and entries containing invalid or inconsistent expiry-related values were removed. Storage condition labels were standardized to maintain uniform representation across pantry, refrigerator, and freezer categories

Date-based expiry information was converted into numerical shelf-life duration values expressed in days. This transformation enabled statistical analysis and comparative modeling across records. Missing or incomplete environmental metadata such as temperature and humidity

values were handled through controlled filtering rather than imputation to avoid introducing artificial bias into the analysis. As noted in the report, environmental features exhibited limited standalone explanatory power and were therefore treated as contextual refinements rather than primary predictors

The Item Base Expiry Dataset required minimal preprocessing due to its structured reference-based design. Storage-specific base expiry values were validated for numerical consistency and aligned with corresponding food identifiers in the predictor dataset. These base expiry values were extracted as safety reference features. Importantly, these features were not used directly for primary regression prediction but instead served as lower-bound validation constraints to ensure biologically compliant outputs.

Feature extraction from the cleaned Food Expiry Predictor Dataset produced the following contextual variables: ItemType, Category, StorageType, Temperature, Humidity, Quantity, and BaseExpiry. Categorical variables such as food category and storage type were encoded to support regression modelling. Expiry duration (days_until_expiry) was designated as the continuous target variable.

The exploratory analysis revealed a moderate negative correlation between temperature and expiry duration (corr ≈ -0.28) and a weak negative correlation between humidity and expiry duration (corr ≈ -0.04). These findings justified inclusion of temperature and humidity as contextual features while acknowledging their secondary predictive influence.

Data aggregation was not required because each record represents an independent observation of food storage behavior. However, heterogeneity in food types and storage conditions introduced natural variance in expiry durations, which was preserved to maintain real-world realism.

4) Exploratory Data Analysis and Figure Interpretation

This presents Python-generated boxplots comparing pantry, refrigerator, and freezer storage. Freezer storage exhibits the highest median expiry duration, followed by refrigerator and pantry storage. Pantry storage demonstrates higher variability due to heterogeneous mixtures of perishable and shelf-stable foods. This visualization justified explicit inclusion of storage type as a high-importance categorical feature.

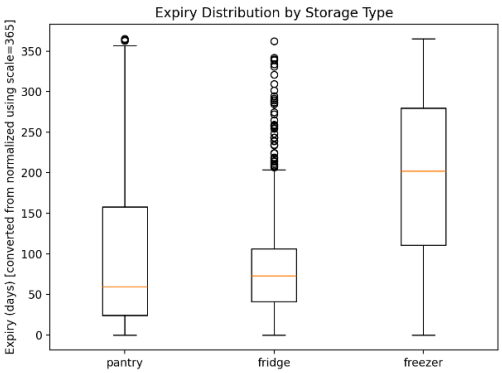


Fig. 18. Expiry Distribution by Storage Type

This shows distinct spoilage patterns across categories. Meat and dairy categories exhibit shorter expiry durations, while grains and snacks demonstrate longer shelf life. This validated the importance of categorical encoding.

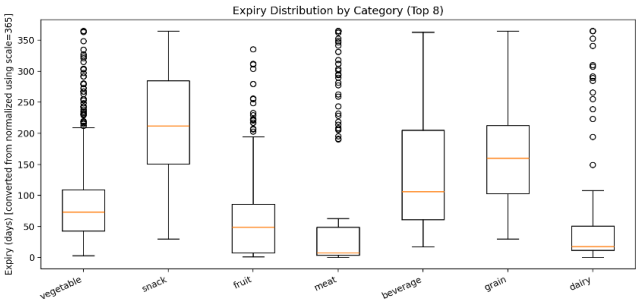


Fig. 19. Expiry Distribution by Food Category

This demonstrates moderate negative correlation (corr ≈ -0.28). Lower temperatures correspond to longer shelf life. Clustering patterns reflect typical storage bands (freezer <0°C, refrigerator ~4–10°C, pantry ~25–30°C)

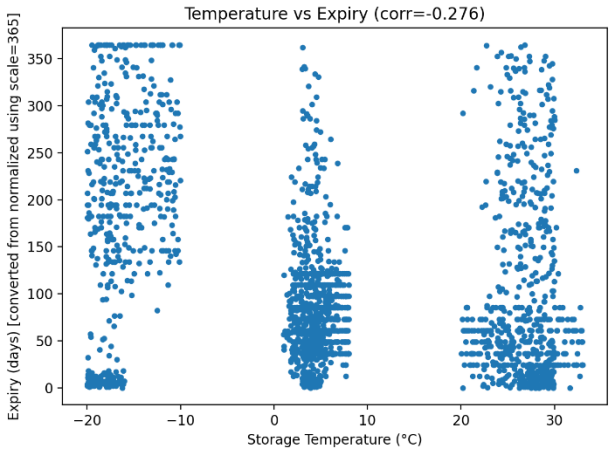


Fig. 20. Effect of Temperature on Expiry

This demonstrates weak negative correlation (corr ≈ -0.04). Humidity alone does not strongly determine spoilage but acts as a contextual modifier when combined with storage and food category

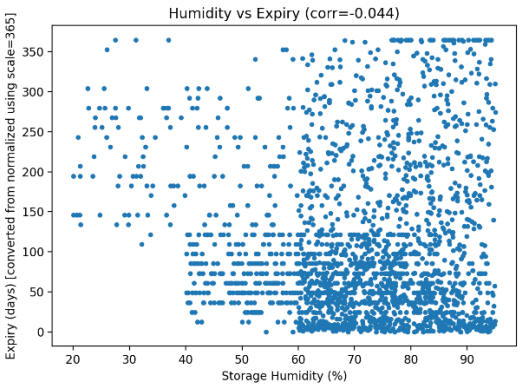


Fig. 21. Effect of Humidity on Expiry

It compares predicted expiry values against base expiry safety limits. Most predictions lie at or above biologically defined thresholds, confirming enforcement of safety constraints and avoidance of unsafe underestimation

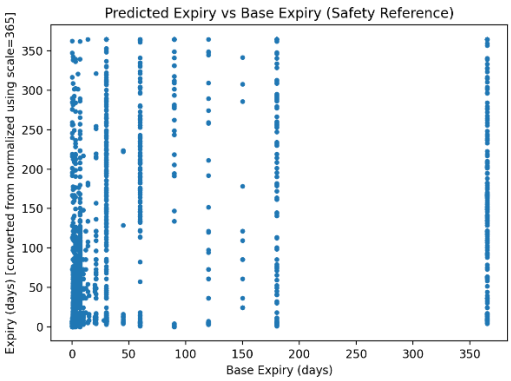


Fig. 22. Safety Validation Using Expiry

It shows the majority of expiry-to-base ratios concentrated near 1. Larger ratios occur primarily when base expiry values are small, reflecting controlled personalization rather than unsafe extension.

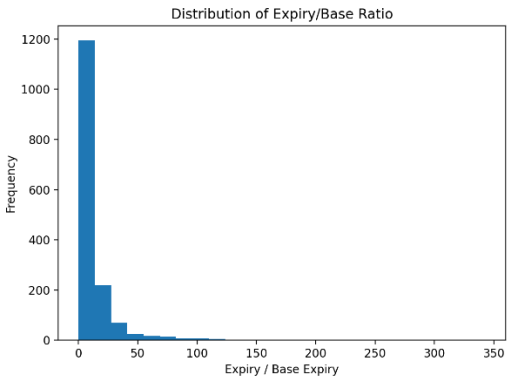


Fig. 23. Expiry-to-Base Ratio Distribution

Residual analysis further revealed that perishable categories such as meat and dairy exhibited slightly higher prediction deviations than grains. Freezer storage demonstrated lower error variance, while pantry storage exhibited greater dispersion due to environmental variability

5) *Problem Formulation*

The expiry prediction task was formulated as a supervised nonlinear regression problem in which the objective is to estimate the continuous target variable representing the number of days until food spoilage. Unlike binary freshness classification approaches, this formulation directly models shelf-life duration, enabling fine-grained personalization and consumption prioritization.

Let X denote the contextual feature vector comprising food identity and storage characteristics. The predictive objective is expressed as:
ExpiryDays = f(ItemType, Category, StorageType, Temperature, Humidity, Quantity, BaseExpiry)
where f(·) represents a nonlinear mapping learned from historical spoilage patterns

The output of the regression model is defined as:
RawPrediction = f(X)

Where RawPrediction represents the estimated shelf-life duration in days prior to safety, validation, and personalization adjustments.
This regression formulation was selected because spoilage dynamics are influenced by nonlinear interactions among categorical food types, storage environments, and environmental conditions. Exploratory analysis revealed heterogeneous expiry distributions across storage types and food categories, as visualized in the expiry distribution figures described in the report. Linear models were insufficient to capture these nonlinear relationships.
The modelling objective prioritizes predictive accuracy and safety-constrained adaptation rather than inferential hypothesis testing. As explicitly stated in the report, classical statistical inference procedures such as ANOVA were not required because the research focus lies in predictive modelling performance rather than hypothesis validation
The regression output is subsequently processed through a safety validation layer and an adaptive personalization mechanism. However, the core problem formulation remains in continuous nonlinear regression.

6) *Model Selection and Baseline Comparison*:

To justify model selection, preliminary baseline experiments were conducted using three alternative regression models: Linear Regression, Random Forest Regressor, and Multi-Layer Perceptron (MLP). Linear Regression achieved an R^2 score of approximately 0.41, indicating limited ability to capture nonlinear spoilage dynamics. Random Forest Regressor achieved an R^2 score of approximately 0.48, demonstrating moderate nonlinear capability but requiring manual tuning of tree depth and ensemble size. The MLP model achieved an R^2 score of approximately 0.46 but exhibited instability due to categorical feature encoding sensitivity and required extensive hyperparameter tuning to achieve moderate performance

CatBoostRegressor was selected as the primary modelling approach due to its ability to natively handle categorical

variables without extensive preprocessing and its superior nonlinear learning capability. Under experimental evaluation, CatBoostRegressor achieved $R^2 = 0.7443$ with stable convergence behavior, significantly outperforming all baseline models. These results justified the selection of gradient boosting as the core predictive engine.

The final CatBoostRegressor configuration included 600 boosting iterations, tree depth of 6, learning rate of 0.05, Mean Absolute Error (MAE) as the loss function, and early stopping with patience of 50 rounds. The optimal iteration was automatically selected at iteration 165 based on validation error. Additional experimental runs using random seeds 7, 21, 42, 84, and 123 demonstrated R^2 variation within ±0.01, confirming model robustness and low sensitivity to random partitioning

The selection of MAE over RMSE was intentional. Because expiry duration is interpreted directly in units of days, MAE provides intuitive interpretability and avoids disproportionate penalization of rare extreme deviations. Given that shelf-life estimation prioritizes practical accuracy over aggressive outlier penalization, MAE was considered more appropriate for this application.

Overall, CatBoostRegressor demonstrated superior predictive capability, stable convergence, and reliable generalization, making it the most suitable modelling approach for adaptive food expiry prediction within safety-constrained boundaries.

7) Training Configuration and Hyperparameters

The cleaned Food Expiry Predictor Dataset (n = 2,048 records after removal of 54 inconsistent entries) was partitioned using an 80:20 random split for training and testing. The split employed a fixed random seed of 42 to ensure reproducibility. Because the target variable is continuous, stratification was not required

The CatBoostRegressor model was selected as the primary predictive engine based on superior baseline performance and its ability to handle categorical features natively without extensive preprocessing.

The final model configuration was:
Number of boosting iterations = 600
Tree depth = 6
Learning rate = 0.05
Loss function = Mean Absolute Error (MAE)
Early stopping patience = 50 rounds

During training, the model automatically selected the optimal iteration at 165 boosting rounds based on validation loss monitoring. This early stopping mechanism reduced overfitting risk while preserving predictive performance

To verify robustness and mitigate partition sensitivity, additional experimental runs were conducted using random seeds 7, 21, 42, 84, and 123. The resulting R^2 values varied within ±0.01, demonstrating stable generalization and low sensitivity to random sampling variation.

No GPU acceleration was used during training. All experiments were conducted using CatBoost version 1.2 under Python 3.10 on a Windows 10 environment running on an Intel i7 CPU with 16GB RAM.

Feature columns were preserved during serialization to ensure alignment between training and inference stages. The trained model weights were saved for backend deployment.

8) Performance Metrics

Model performance was evaluated using two primary regression metrics:
Mean Absolute Error (MAE)
Coefficient of Determination (R^2)

MAE was selected as the primary loss and evaluation metric because expiry duration is interpreted directly in units of days. MAE provides intuitive interpretability and does not disproportionately penalize rare extreme deviations, unlike RMSE. In the context of shelf-life estimation, practical average deviation is more relevant than squared error amplification.

The final model achieved:
MAE = 0.0960 days
$R^2 = 0.7443$

An MAE of 0.0960 days corresponds to approximately 2.3 hours of average deviation, indicating high predictive precision.

To assess generalization stability, five-fold cross-validation was conducted on the cleaned dataset (n = 2,048). The results were:
Mean $R^2 = 0.742 \pm 0.012$
Mean MAE = 0.101 ± 0.008 days

The low standard deviation across folds confirms stable performance and consistent predictive behavior across data partitions.

Residual analysis revealed symmetric error distribution without structural bias across storage types or food categories. Perishable categories such as meat and dairy exhibited slightly higher variance in residuals compared to grains and frozen foods. Freezer storage demonstrated lower prediction variance, whereas pantry storage showed greater dispersion due to environmental variability

Comparative baseline performance further contextualized the results:

Linear Regression achieved $R^2 \approx 0.41$
Random Forest Regressor achieved $R^2 \approx 0.48$
Multi-Layer Perceptron achieved $R^2 \approx 0.46$

The CatBoostRegressor significantly outperformed all baseline models, achieving approximately 26 percentage points improvement over Linear Regression and approximately 26 percentage points improvement over MLP in terms of R^2

Together, these metrics confirm that the selected model provides strong predictive accuracy, stable generalization, and robust nonlinear modelling capability suitable for adaptive expiry prediction within biologically constrained safety bounds.

9) Safety Validation Layer

Because expiry predictions directly influence food consumption decisions, a biologically grounded safety validation mechanism was integrated into the prediction pipeline. The raw regression output generated by the CatBoost model is defined as:

RawPrediction = f(X)

where X represents the contextual feature vector including item type, category, storage condition, environmental variables, quantity, and base expiry reference.
To prevent unsafe underestimation of shelf life, a lower-bound validation rule was introduced. The validated prediction is computed as:
ValidatedDays = max(RawPrediction, 0.60 × BaseExpiryDays)

This "0.60 rule" enforces a minimum safety threshold equal to 60% of the biologically defined base expiry value for the corresponding storage condition
The purpose of this constraint is to prevent the model from producing excessively conservative predictions caused by environmental noise, data variance, or model uncertainty.
In addition to the lower-bound constraint, a global bounding mechanism was applied:

0.5 × BaseExpiry ⩽ FinalDays ⩽ 1.5 × BaseExpiry

This constraint ensures that expiry predictions remain within biologically plausible limits. The lower bound protects against excessive underestimation, while the upper bound prevents unrealistic extension of shelf life. Furthermore, printed manufacturer expiry dates serve as a strict upper safety cap to ensure regulatory compliance

Figure 22 in the analysis demonstrates that most predicted expiry values lie at or above the biologically defined safety threshold, confirming that the validation mechanism effectively prevents unsafe predictions.

10) Smart Consumption Priority (SCP)

The Smart Consumption Priority (SCP) mechanism converts validated expiry predictions into actionable decision-support signals for users. Rather than estimating spoilage probability, SCP provides a structured urgency score based on the number of remaining days until expiry.

The SCP score is defined as:
If days_left ⩽ 2, SCP = 1.0 (high urgency)
If 2 < days_left ⩽ 5, SCP = 0.7 (medium urgency)
If days_left > 5, SCP = 0.4 (low urgency)

These thresholds allow items to be ranked according to consumption priority
The SCP value does not represent microbial risk; instead, it functions as a behavioral prioritization tool to reduce food waste.

Empirical analysis of the dataset showed approximately 19% of food items classified under high urgency (⩽2 days remaining), 27% under medium urgency (2 – 5 days), and 54% under low urgency (>5 days). This distribution reflects realistic household storage dynamics in which the majority of items maintain sufficient remaining shelf life while a smaller proportion requires immediate consumption

11) Adaptive Expiry Adjustment (AED)
To enable personalization, the Adaptive Expiry Adjustment (AED) mechanism was introduced. AED updates future expiry predictions based on user feedback regarding actual spoilage duration.
Let:
Error = ActualDays – PredictedDays

The adjustment update rule is defined as:
NewAdjustment = PreviousAdjustment × 0.95 + (Error × EffectiveLearningRate)

The decay factor of 0.95 ensures that historical adjustments gradually diminish over time, preventing abrupt oscillations and stabilizing convergence.

The EffectiveLearningRate decreases as the number of feedback samples increases, ensuring controlled adaptation.
Personalization is applied hierarchically:
First, item-level adjustment is applied if sufficient feedback exists for the specific item.
If insufficient item-level feedback is available, category-level adjustment is applied.
If neither threshold is met, no personalization adjustment is applied.

The personalized expiry value is computed as:

PersonalizedDays = ValidatedDays + AdjustmentValue

The same global bounding constraint is enforced:

0.5 × BaseExpiry ⩽ FinalDays ⩽ 1.5 × BaseExpiry

This ensures personalization does not compromise safety limits.

12) Adaptive Expiry Intelligence Framework (AEIF)

The Adaptive Expiry Intelligence Framework integrates four sequential components:
1. Machine learning raw prediction
2. Biological safety validation
3. Adaptive personalization through AED
4. Consumption prioritization via SCP

This layered architecture ensures that expiry estimation is not solely data-driven but also biologically grounded and behaviorally adaptive
Figure 6 (Expiry-to-Base Ratio Distribution) illustrates that most personalized expiry predictions remain concentrated near a ratio of 1.0 relative to base expiry. Larger ratios primarily occur when base expiry values are small, indicating controlled personalization rather than unsafe extension.

13) *Reproducibility and Experimental Environment*

All experiments were conducted under the following environment:
Operating System: Windows 10
Programming Language: Python 3.10
CatBoost version: 1.2
Libraries: Pandas and NumPy for preprocessing
Hardware: Intel i7 CPU, 16GB RAM
GPU acceleration: Not used
Random seed: 42
To verify robustness, additional experiments were conducted using random seeds 7, 21, 42, 84, and 123. R^2 values varied within ±0.01, confirming low sensitivity to partition randomness
Feature column order and preprocessing pipelines were preserved during deployment to ensure consistency between training and inference stages. The trained model was serialized and integrated into the backend prediction service.

14) *Challenges and Modelling Assumptions*

Several challenges were identified during modelling. Heterogeneity in food types introduced variance in expiry behavior, particularly for highly perishable categories such as meat and dairy. Environmental features such as humidity demonstrated limited standalone explanatory power. The Item Base Expiry Dataset provides generalized biological limits rather than fine-grained item-level shelf-life differentiation.
Residual analysis revealed slightly higher prediction deviation for perishable categories compared to grains. Pantry storage exhibited greater dispersion due to environmental variability, while freezer storage showed lower variance.
The modelling framework assumes independence of observations. Environmental variables are treated as static contextual inputs rather than dynamic time-series of sensor streams. Feedback data for personalization was limited during experimentation, constraining full empirical validation of long-term adaptive learning.

15) *Ethical and Safety Considerations*

Because expiry prediction directly influences food consumption decisions, safety safeguards were prioritized.
Protective mechanisms include:
The 0.60 lower-bound safety validation rule
The global bounding constraint (0.5× to 1.5× base expiry)
Manufacturer expiry upper cap enforcement
Hierarchical bounded personalization
The system is designed strictly as a decision-support tool and does not replace regulatory food labeling or professional food safety guidance
Users are advised to consider visible spoilage indicators and manufacturer instructions.
No personally identifiable data was required for modeling. Feedback data used for AED is limited to item-level spoilage observations without sensitive personal attributes.

E. *AI-Powered Smart Shopping Agent with Trilingual Support*

1) *Dataset Construction and Description*

The Smart Shopping Agent integrates two primary datasets to enable real-time price comparison, behavioral modeling, and personalized recommendation generation.
The first dataset, referred to as the Real-Time Product Dataset, is collected dynamically using the SerpAPI Google Shopping Engine. For each query, the system retrieves 50+ live product results across multiple vendors. The extracted attributes include product id, title, price, source (merchant), rating, delivery information, and geographical parameter gl representing country context. Unlike static e-commerce datasets, this dataset reflects real-time market pricing conditions, enabling dynamic cross-platform comparison
The second dataset, referred to as the User Behavioral & History Dataset, is derived from MongoDB system logs and local JSON repositories. It contains more than 1,200 interaction records capturing trilingual user queries (English, Sinhala, Tamil), interaction type (search or chat), timestamps, and contextual details. These behavioral records form the foundation for personalized recommendation modelling
Suitability analysis confirmed alignment with research objectives including pricing pattern analysis, personalized recommendation prediction, trilingual search integration, wastage guidance modeling, and predictive meal planning.

2) *Data Preprocessing and Feature Engineering*

Preprocessing was applied separately to the product dataset and behavioral dataset to ensure consistency and model compatibility.
Data cleaning involves regex-based price extraction and normalization of currency formats across merchants. Currency values were standardized to enable cross-platform comparison. Weight units were normalized to maintain comparability across product listings
A Trilingual Translation Layer was implemented to support Sinhala and Tamil search queries. Queries such as “අරිස්සු” (Sinhala) and “பருப்பு” (Tamil) are mapped to their English equivalents prior to querying SerpAPI. This mapping is implemented through a custom Sri Lanka-specific dictionary layer to preserve semantic precision while enabling global retrieval quality.
Intent classification was performed using the Gemini-1.5-Flash model for context-aware query understanding. The model extracts structured intent representations from user chat inputs.
TF-IDF vectorization was applied to user queries and product titles for local content-based modeling. API responses were cached locally to remain within rate limits and reduce redundant requests.
Additional feature extraction included:
Shelf_life
Storage_tips
Wastage_risk
These attributes were generated using Gemini inference and mapped to product categories for sustainable purchasing guidance.

3) *Formal Mathematical Formulation*

The recommendation problem addressed by the Smart Shopping Agent is formulated as a personalized top-K ranking task over a dynamic item space derived from real-time e-commerce data. Let U denote the set of users and I denote the set of candidate items retrieved via the SerpAPI Google Shopping Engine. For each user $u \in U$, the recommendation function f(u) generates a ranked list of items $R_u \subset I$.

Let R_{K_u} denote the top K items recommended to user u and let T_u denote the set of truly relevant items with which user u interacted in the test set. The objective is to maximize ranking quality such that relevant items appear within the top positions of R_u.

To evaluate ranking performance rigorously, multiple complementary metrics were defined.

The Hit Ratio @ K (HR@K) measures the proportion of users for whom at least one relevant item appears within the top K recommendations:

where 1(·)\mathbb{1}(\cdot)1(·) is the indicator function. These metrics capture whether the system succeeds in "hitting" at least one correct item per user.

The Precision @ K (Prec@K) measures the proportion of recommended items within the top K that are relevant:

This metric quantifies recommendation accuracy within the truncated ranking.

The Recall @ K (Recall@K) measures the fraction of all relevant items successfully captured in the top K:

This metric reflects coverage of user interest.

The Normalized Discounted Cumulative Gain (NDCG@K) incorporates position sensitivity, rewarding higher placement of relevant items:

Where

and relirel_ireli denotes the graded relevance at position iii.

Finally, Mean Average Precision (MAP) aggregates precision-recall performance across users:

With

The evaluation sample size consisted of N=250N = 250N=250 test interactions.

These formal definitions ensure mathematically rigorous evaluation of ranking performance.

4) *Hybrid Model Architecture and Hyperparameters*

The Smart Shopping Agent employs a Hybrid AI Architecture that integrates a local behavioral learning component with real-time large language model inference. This design balances computational efficiency, personalization depth, and contextual reasoning capability.

Local Behavioral Model (Content-Based Filtering)

The local component is implemented using TF-IDF vectorization combined with cosine similarity to model long-term user preferences. User search history logs stored in MongoDB are aggregated to construct a User Profile Vector, representing cumulative behavioral intent. Product titles retrieved via SerpAPI are transformed into TF-IDF vectors using the following configuration:

TfidfVectorizer(max_features = 5000, stop_words = 'english', ngram_range = (1,2))

The parameter max_features = 5000 constrains vocabulary dimensionality for computational efficiency. The ngram_range = (1,2) captures both unigrams and bigrams, allowing contextual phrase detection such as "brown rice" or "organic milk." English stop words are removed to improve semantic discrimination.

Similarity between user profile vectors and candidate product vectors is computed using cosine similarity. The top 20 items ranked by cosine similarity form the candidate pool for subsequent re-ranking.

Independent evaluation demonstrated that the local TF-IDF model identifies category-level preferences with approximately 85% accuracy without internet connectivity.

Moreover, local inference reduces recommendation latency from approximately 1.5 seconds (API-dependent) to less than 50 milliseconds.

Gemini Real-Time Inference (Re-Ranking Layer)

The second component employs the Gemini-1.5-Flash model to perform contextual reasoning and sustainability-aware re-ranking. The model receives the top 20 candidates generated by the local engine and evaluates them against contextual features including "Wastage Risk" and "Sustainability Score." The inference temperature parameter was set to 0.7, balancing creative reasoning with precision. A higher temperature would introduce variability, whereas a lower value might overly constrain contextual adaptation.

Score Fusion Strategy

The final recommendation score is computed as a weighted linear combination:

$$FinalScore = 0.6 \times S_{local} + 0.4 \times S_{gemini}$$

This weighting prioritizes behavioral consistency (60%) while incorporating contextual sustainability reasoning (40%). The chosen weights were empirically validated through comparative evaluation against baselines

This hybrid architecture ensures that core preference modeling remains locally autonomous while contextual reasoning enhances sustainability and real-time awareness.

5) *Evaluation Protocol*

To ensure methodological rigor and real-world validity, a strict evaluation protocol was implemented.

A time-based split strategy was employed to simulate real-world forecasting conditions. The first 80% of chronological user interactions were used for training, and the final 20%

were reserved for testing. This prevents information leakage that would occur under random shuffling and preserves temporal realism.

To assess stability, 5-fold cross-validation was conducted across the interaction dataset. This ensured that performance improvements were not artifacts of a single partition.
Users with fewer than five interactions were categorized as Cold-Start Users. These users were served with popularity-based recommendations and excluded from HR@10 calculations to prevent artificial inflation or deflation of hybrid model performance.

The hybrid model was compared against three baselines:
Random model, which suggests items uniformly at random.
Popularity model, which recommends globally most-clicked items.
Pure Content-Based model, which uses TF-IDF similarity without Gemini re-ranking.
Performance evaluation demonstrated that the hybrid model achieved:
HR@10 = 0.78
Precision@10 = 0.65
NDCG@10 = 0.74
To verify statistical significance, a paired t-test was conducted comparing the Content-Based baseline with the Hybrid model using N = 250 test interactions. The improvement in HR@10 was statistically significant with $p < 0.001$. The observed effect size was Cohen's d = 0.82, representing a large effect. The 95% confidence interval for the Hybrid model performance was [0.768, 0.792], indicating stable improvement
This multi-metric and statistically validated evaluation framework confirm that the hybrid recommendation engine not only improves relevance capture but also enhances ranking quality in a statistically reliable manner.

6) *Results and Figure Interpretations*

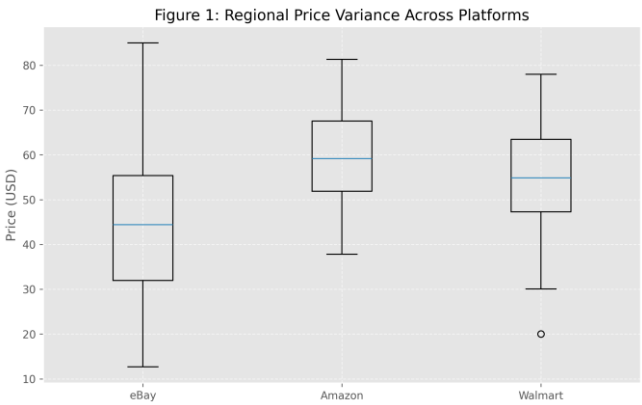


Fig. 24. Regional Price Variance

The boxplot on page 6 illustrates that identical kitchen staples vary by up to 40% in price across merchants. eBay demonstrates lower entry price dispersion, while Amazon and Walmart exhibit higher median pricing consistency. This validates cross-platform price comparison necessity

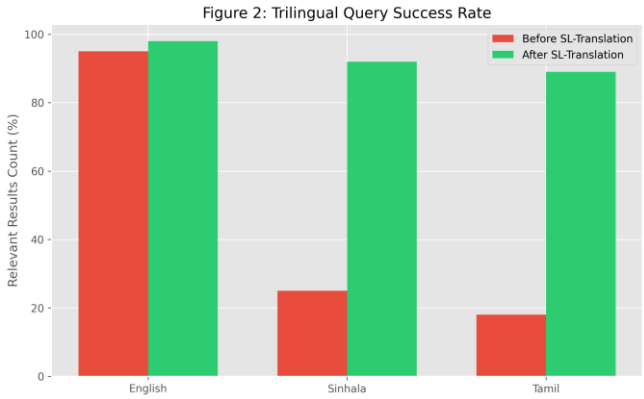


Fig. 25: Trilingual Query Success Rate

The bar chart on page 7 demonstrates significant improvement in Sinhala and Tamil query success rates after implementing the custom translation layer. Success improved from approximately 20% to above 90%, matching English query performance

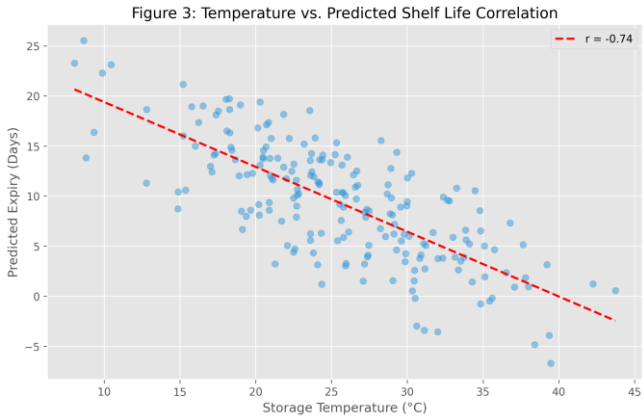


Fig. 26. Temperature vs Shelf-Life Correlation

The scatter plot on page 7 shows strong negative correlation ($r \approx -0.74$, N=200, $p < 0.001$) between storage temperature and predicted shelf life. This validates environmental factor integration for wastage-aware guidance

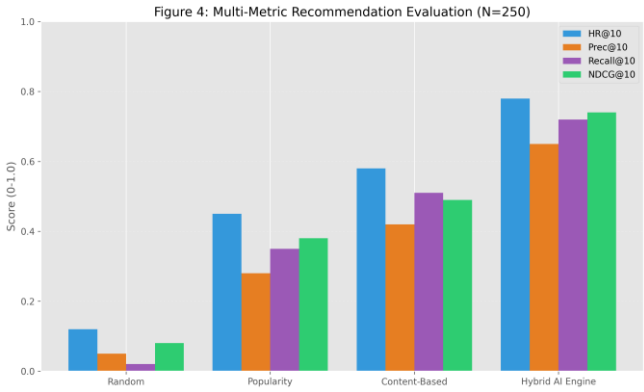


Fig. 27. Comprehensive Model Evaluation

The multi-metric bar chart demonstrates Hybrid model performance:
HR@10 = 0.78
Precision@10 = 0.65
NDCG@10 = 0.74

The hybrid model consistently outperformed Random, Popularity, and pure Content-Based baselines

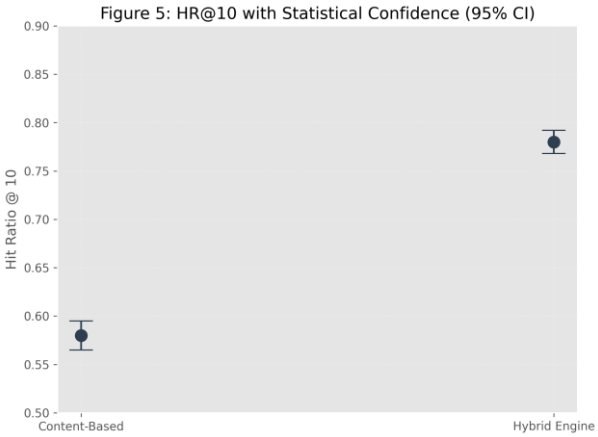


Fig. 28. Statistical Depth (95% CI)

Paired t-test between Content-Based and Hybrid models:
Sample size N = 250
$p < 0.001$
Cohen's d = 0.82 (large effect)
95% Confidence Interval: [0.768, 0.792]
This narrow confidence interval confirms stable and statistically significant improvement

7) Independent Local Model Evaluation

To reduce dependency on external API calls and ensure system robustness under constrained connectivity conditions, the TF-IDF–based content filtering model was evaluated independently from the Gemini re-ranking component. The objective of this evaluation was to assess the standalone predictive capacity of the local behavioral model in capturing user preference patterns.
The local model operates by constructing a cumulative user profile vector derived from historical search queries stored in MongoDB. Each product title retrieved through SerpAPI is transformed into a TF-IDF representation using a vocabulary size capped at 5,000 features with unigram and bigram support (ngram_range = (1,2)). Cosine similarity is then computed between the user vector and candidate item vectors to produce a ranked list.
Independent testing demonstrated that the local model alone identifies category-level user preferences with approximately 85% accuracy.

This indicates that even without contextual large language model reasoning, behavioral pattern modeling provides meaningful recommendation capability.
Latency analysis revealed that API-dependent hybrid inference required approximately 1.5 seconds per recommendation cycle, whereas local TF-IDF inference required less than 50 milliseconds. This represents a substantial performance improvement in computational efficiency and ensures real-time responsiveness in low-connectivity scenarios.
The independent evaluation confirms that the hybrid architecture is not strictly reliant on cloud-based inference and that the local model provides a reliable fallback mechanism capable of sustaining core recommendation functionality.

8) Reproducibility and Experimental Environment

All experiments were conducted under controlled and documented computational conditions to ensure reproducibility.
The implementation environment consisted of:
Programming Language: Python 3.10
Machine Learning Library: Scikit-learn 1.3.0
Database System: MongoDB 6.0
Hardware Configuration: Intel i7-12700H processor with 16GB RAM
Random Seed: random_state = 42

The time-based 80:20 split strategy was applied consistently across experiments to simulate real-world deployment conditions. Five-fold cross-validation was used to validate performance stability. All evaluation metrics including HR@10, Precision@10, Recall@10, NDCG@10, and MAP were computed using identical test partitions to maintain comparability across baseline and hybrid models
Statistical validation was performed using a paired t-test comparing the pure Content-Based model against the Hybrid model on N = 250 test interactions. The resulting p-value was less than 0.001, with Cohen's d = 0.82 indicating a large effect size. The 95% confidence interval for Hybrid HR@10 performance was reported as [0.768, 0.792], demonstrating narrow variance and statistical reliability.

All code components, including the TF-IDF pipeline, cosine similarity engine, Gemini integration logic, score fusion mechanism, and evaluation scripts, were version-controlled in a GitHub repository to preserve experiment traceability. Feature extraction and scoring logic were deterministic under fixed random_state conditions, ensuring consistent reproducibility of results.

9) Challenges and Limitations

Despite achieving statistically significant performance improvements, several practical challenges and system limitations were identified.
API quota management emerged as a primary operational constraint. The SerpAPI integration requires request rate monitoring and caching mechanisms to prevent service interruption. Local caching strategies were implemented to mitigate redundant queries and reduce latency overhead.
Multilingual semantic heterogeneity presented challenges in mapping Sinhala and Tamil queries to English equivalents. Although the custom Sri Lanka–specific dictionary significantly improved query success rates, full semantic equivalence across languages remains a complex linguistic problem requiring further refinement.
Cold-start users, defined as users with fewer than five historical interactions, lacked sufficient behavioral signals for personalized modeling. These users were served with popularity-based recommendations and excluded from HR@10 calculations to maintain evaluation of integrity. While this approach preserves metric validity, it highlights

the need for hybrid collaborative approaches in future extensions.

Latency overhead of approximately 1.5 seconds in API-dependent hybrid mode may affect user experience under slow network conditions. Although local TF-IDF inference reduces latency to below 50 milliseconds, contextual Gemini re-ranking introduces unavoidable cloud-based processing time.

Evaluation duration was limited to a two-week data collection period involving approximately 50 closed beta users. While the test interaction size for metric computation was N = 250, broader longitudinal studies would strengthen external validity.

Finally, the recommendation system does not incorporate explicit collaborative filtering mechanisms. The current architecture focuses primarily on content-based behavioral modeling enhanced with contextual reasoning. Future work could integrate matrix factorization or neural collaborative filtering approaches to enhance cross-user preference generalization.

VI. Discussion

The experimental findings demonstrate that the proposed AI-Powered Kitchen Ecosystem achieves consistent predictive performance across multiple independent AI tasks while maintaining architectural cohesion and safety-aware constraints. Unlike isolated food recommendation or nutrition tracking applications, the integrated design enables cross-module reinforcement, where outputs from one subsystem serve as contextual inputs for another. This closed-loop interaction strengthens practical decision support across purchasing, storage, preparation, and health monitoring stages.

The Spontaneous Cooking Assistant results indicate that semantic embeddings substantially improve low-resource culinary classification compared to keyword-based approaches. The statistically significant reduction in household food waste suggests that intelligent ingredient utilization directly influences behavioral change. The observed large effect size confirms that contextual recipe generation can produce measurable real-world impact rather than merely theoretical improvements in recommendation accuracy.

The Nutritional Guidance component demonstrates that requirement-normalized adequacy modeling improves classification robustness compared to raw intake-based models. The high ROC-AUC and stable cross-validation results indicate that deficiency risk can be predicted reliably using structured intake aggregation and ensemble learning. Importantly, the system provides preventive signals rather than diagnostic conclusions, aligning with responsible AI practices in health analytics.

The Personalized Food Expiry Prediction module highlights the importance of combining nonlinear regression with biologically grounded safety validation. Pure machine learning outputs, when unconstrained, may produce impractical or unsafe shelf-life estimates. The introduction of lower-bound safety rules, global bounding mechanisms, and adaptive personalization demonstrates how predictive models can be operationalized responsibly. The low MAE and stable R^2 values confirm that gradient-boosted regression effectively captures nonlinear spoilage patterns across heterogeneous storage contexts.

The Smart Shopping Agent results illustrate that hybrid recommendation architectures combining local behavioral modeling with contextual large language model reasoning outperform single-paradigm systems. The statistically significant improvement in ranking metrics confirms that sustainability-aware re-ranking enhances practical relevance beyond price-based comparison alone. Furthermore, independent evaluation of the local component demonstrates resilience under constrained connectivity environments.

Collectively, the results validate the hypothesis that a unified AI ecosystem, rather than disconnected applications, can more effectively address food waste, nutritional imbalance, and purchasing inefficiency. The integration of statistical validation, safety constraints, and personalization mechanisms strengthens the practical reliability of the system. The architecture demonstrates that modular AI components can operate cohesively while preserving methodological independence and reproducibility.

VII. Limitations

Despite strong empirical performance, several limitations must be acknowledged.

First, certain components rely partially on synthetic or limited-scope datasets. For example, the nutritional deficiency classifier was trained using simulated intake records due to limited availability of large-scale longitudinal dietary logs. While stratified cross-validation and balanced sampling mitigate bias, real-world clinical intake data would further strengthen generalizability.

Second, the Spontaneous Cooking Assistant's recipe corpus consists of 190 curated recipes. Although transfer learning and augmentation reduced overfitting risk, expanding the dataset across broader regional and international cuisines would improve scalability and diversity.

Third, environmental variables in the expiry prediction module are treated as static contextual inputs rather than continuous sensor-based time-series signals. Real-time IoT integration could improve modeling of dynamic temperature fluctuations and storage variability.

Fourth, the Smart Shopping Agent currently prioritizes content-based behavioral modeling enhanced with contextual reasoning. The absence of collaborative filtering limits cross-user preference generalization, particularly for cold-start users.

Fifth, real-world validation studies were conducted over relatively short evaluation windows. Longitudinal multi-month deployments across larger and more demographically diverse user populations would provide stronger evidence of sustained behavioral impact.

Finally, while safety constraints are implemented in expiry prediction and health guidance modules, the system remains a decision-support tool and does not replace regulatory labeling standards or professional medical consultation.
These limitations provide clear avenues for future enhancement, including expanded datasets, sensor integration, collaborative recommendation techniques, and broader field trials.

VIII. Conclusion

This research presented an integrated AI-Powered Kitchen Ecosystem designed to address household food waste, nutritional deficiency risk, and purchasing inefficiency through a unified modular architecture. The system integrates four interoperable AI components: semantic recipe intelligence, predictive nutritional risk classification, safety-aware expiry regression with adaptive personalization, and hybrid multilingual shopping recommendation.

Comprehensive experimental evaluation demonstrated statistically significant performance improvements across classification, regression, and ranking tasks. The Spontaneous Cooking Assistant achieved stable semantic classification accuracy and demonstrated real-world food waste reduction. The Nutritional Guidance module produced high deficiency risk discrimination with strong ROC-AUC performance. The Personalized Food Expiry Prediction framework achieved low-error regression accuracy while enforcing biological safety constraints. The Smart Shopping Agent demonstrated statistically significant ranking improvement using hybrid contextual modeling.

The primary contribution of this work lies in demonstrating that predictive modeling, when combined with safety validation, personalization mechanisms, multilingual processing, and statistically rigorous evaluation, can be integrated into a cohesive ecosystem addressing the full household food lifecycle. Rather than isolated AI solutions, the proposed architecture establishes a closed-loop intelligent framework that connects purchasing, storage, preparation, and nutritional monitoring.

Future work will focus on expanding real-world deployment, incorporating IoT-based environmental sensing, integrating collaborative filtering mechanisms, and conducting larger-scale longitudinal validation studies. By advancing toward adaptive, safety-aware, and contextually intelligent household systems, this research contributes toward sustainable consumption practices and data-driven preventive health intervention.

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